

Workload balancing for flight dispatchers

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ARTICLE INFO

Keywords:

Flight dispatcher
Scheduling
Workload balancing
Lagrangian relaxation
Heuristic

ABSTRACT

Flight dispatchers are responsible for flight planning prior to departure and flight monitoring while en-route. Their work involves multi-tasking and their workload is dynamic. We study such a problem under two nonlinear workload balancing measures: minimum peak workload and minimum absolute deviation. In order to solve practical instances efficiently, we use decomposition through Lagrangian relaxation to reduce the problem into easier-to-solve subproblems and prove that the Lagrangian lower bound has a closed-form expression for the peak workload objective. To find feasible solutions, we develop a Focus-Search-and-Improve heuristic with a genetic algorithm core where parts of the feasible solution set are explored and searched by a genetic algorithm, and solutions are further fine-tuned by an improvement heuristic. To test the efficiency of the proposed approach, we generated 231 instances based on 2019 U.S. Bureau of Transportation flight data that involve 17 different carriers and up to 3968 flights per instance. Numerical testing demonstrates the efficiency of the proposed approach in that the Lagrangian lower bound is very tight, and the heuristic finds optimal solutions in 33.4% of the instances and are on average 3.5% away from the Lagrangian lower bound. It also reveals that the difficulty of the problem increases for smaller workstation-to-flight ratios, and that the peak workload objective achieves the goal of balancing the workload at times where peaks occur but does not necessarily balance the workload throughout the workday. On the other hand, the absolute deviation objective achieves better balance between workstations at the expense of a slight increase in peak workload.

1. Introduction

The airline industry is a driving force for global economic growth with around 8.5 billion passengers in 2023 and a yearly demand growth of 13.8% in March 2024, measured in revenue passenger kilometers (ACI, 2024; IATA, 2024). Despite facing difficulties in the past, such as fluctuating fuel prices and economic downturns, the COVID-19 pandemic has been the most severe challenge it has encountered. In an attempt to recover to pre-pandemic levels, airline companies are exploring a multitude of cost saving options, including optimizing operational efficiency.

Commercial aviation is a complex business that requires teams of professionals working around the clock to ensure optimal operations planning and safe and efficient execution. While passengers interact with front-end staff such as flight attendants and pilots, there are behind-the-scene roles crucial to flight operations. One such role is that of flight dispatchers who are responsible for flight planning before departure and flight following when en-route. They make critical decisions on flight operations and share legal responsibility on flight operation with pilots. Hence, they require rigorous training and certification. To obtain a flight dispatcher license in the U.S., one must undergo 200 h of training and is required to pass a knowledge test

in meteorology, navigation, communications, air traffic control federal aviation regulations, aircraft operations, among others (US Code of Federal Regulations Title 14, 2022a).

Flight dispatcher tasks may be broadly categorized into two main phases based on the status of flights: pre-departure planning and en-route following. Before departure, dispatcher planning tasks include: fuel, passenger, and cargo load for efficiency; flight adherence to regulations and standards; and flight path determination and release. In particular, dispatchers plan flight path directions and restrictions and timing constraints. They calculate fuel requirements, decide on loading additional payload, check for weather conditions along the flight path, and determine landing options in case of emergency. The dispatcher compiles this information, along with other technical details, in a dispatch release report, which is then sent to air traffic control at least 30 to 60 min before flight departure (US Code of Federal Regulations Title 14, 2022b). En-route responsibilities during the actual flight include continuous communication with pilots, regular checking for weather along the flight path, and monitoring any potential events that could jeopardize safety. In the event of a contingency, such as a medical emergency, the flight crew will turn to the dispatcher for guidance and decision-making.

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Under high workload conditions, dispatchers may not have sufficient time to make detailed analysis and may resort to conservative approaches, which can have financial implications for airlines. For instance, when making fuel calculations, dispatchers may tend to use intentional overestimates and add extra fuel for safety, reducing the fuel efficiency of the flight. Even though en-route tasks may require less involvement compared to pre-flight planning, dealing with contingencies such as changing weather adds to the dispatcher workload and allows for less time for efficiency planning (Huang et al., 2019). Flight dispatchers are crucial for managing unexpected delays and preventing them from propagating into upcoming flights (Abdelghany et al., 2024).

As noted by Tan et al. (2024), workload management is of great importance in aviation operations, with direct implications for efficiency, safety, and cost. Investments in advanced simulation-based training, stress management programs, and decision-support systems for aviation operations may improve dispatcher workload balance and their ability to keep up with the dynamic environment.

Airline companies have their own flight dispatchers that work out of their operations center. Dispatchers operate at workstations which are desks equipped with the necessary systems such as the flight management system, and meteorological data sources. Workstations are up to 24 h long, and are covered by dispatchers who work in shifts, usually 8 to 12 h long covering the duration the workstation is open. In other words, only one dispatcher works at a workstation at any given time and a workstation may be covered by multiple dispatchers consecutively. In order to ensure smooth handover between dispatchers, there is a shift overlap for consecutive dispatchers. Fig. 1 shows a workstation that is operational from 4:00 am to 10:00 pm and has 10 flights assigned, where a flight is shown by a horizontal line spanning departure to arrival times. The workstation is covered by two dispatchers with shifts of 8.5 and 10 h, respectively and a 30 min overlap for takeover. Being responsible for planning and monitoring multiple flights simultaneously, dispatchers are expected to multi-task, which stresses the role they play in maintaining operational efficiency and the level of workload they experience. Dispatchers operate under fluctuating levels of workload as flights change phases and unforeseen circumstances arise. Their workload depends on the set of flights assigned to their workstation at any given time and can have a significant impact on their performance and the cost and safety of flight operations. This is the motivation for the current work, where we study the allocation of flights to workstations with the purpose of balancing dynamic workload.

Airline operations planning like flight and crew scheduling have substantial bodies of literature devoted to them and share mixed integer optimization models and path-based column generation solution frameworks. More comprehensive models involving integrated decisions, practical features or uncertainty/robustness may lead to larger models that are further tackled with metaheuristics; see for example (Xu et al., 2021) and the review paper (Xu, 2024). In contrast, there is scarce dispatcher related work and the flight assignment problem does not have a path structure that is amenable to column generation like other airline problems due to the fact that flights are assigned to and processed at a workstation simultaneously. Another unique feature is the measure for workload both for individual tasks and for cumulative workload assigned to a single dispatcher. There is very little research on dispatcher workload measurement. In this paper we introduce load as a unit-less quantity that captures the relative effort of flight planning and following as a function of flight distance. This measure allows the use of an additive function of task loads to calculate dispatcher workload. While it falls outside the scope of this work, the authors recognize that quantifying dispatcher workload is a research question by itself and is crucial to workload balancing. We note that the problem under study may seem to have similarity with other known problems which we explore in the literature review and show that the problem is unique and novel.

The contributions include the definition of a new class of problems, the solution methodology that can handle real-life problems, a test set and detailed testing. We define the flight dispatcher workload balancing problem that assigns flights to workstations so that the workload imbalance amongst the workstations over the operational time is minimized. We examine the problem under two measures for workload balance: peak workload and absolute deviation from the average workload. This problem fills a gap in the literature on a critical airline operations planning problem and establishes a new class of problems on task assignment to multiple agents with multi-tasking. We provide a mathematical programming formulation and use Lagrangian relaxation to decompose the problem into easier-to-solve subproblems and provide a lower bound. We prove that the Lagrangian lower bound has a closed-form expression, removing the need to iteratively search for it through subgradient and/or column generation type of methods. To find feasible solutions and guided by the Lagrangian lower bound, we develop a Focus-Search-and-Improve metaheuristic with a Genetic Algorithm core (FSI-GA) where restrictions of the full problem are explored and searched by a genetic algorithm, and solutions are further fine-tuned by an improvement heuristic. We performed extensive testing on 2019 U.S. historical domestic flight data from the Bureau of Transportation Statistics (Bureau of Transportation Statistics, 2023b) and analyzed solution quality, computational time, and workload balance metrics. For the peak workload objective, we found that the commercial solvers fail for problem sizes as small as 750 flights and 15 workstations. The Lagrangian lower bound is optimal in all instances with proven solution, which corresponds to 40% of instances, and the (FSI-GA) heuristic is on average 3.1% away from the Lagrangian bound. The peak workload objective achieves the goal of balancing the workload at times where peaks occur but does not directly balance the workload during other times. On the other hand, the absolute deviation objective achieves better workload balance, but it is more difficult to solve.

The paper is organized as follows: In Section 2, the relevant literature on flight dispatching and workload balancing is reviewed. In Section 3, the problem setting and mathematical models for two objectives are introduced. In Section 4, a Lagrangian relaxation is proposed and optimal Lagrange bound is derived for the peak workload objective. In Section 5, a genetic algorithm-based search heuristic is proposed. In Section 6, problem instances are constructed based on real world data, and numerical results are analyzed. Finally, in Section 7, conclusions and future research directions are provided.

2. Literature review

Despite the rich literature on airline operations planning, we only found three studies that focus on flight dispatcher scheduling and assignment. This does not speak to the importance of the problem, but rather raises a question to whether the problem is entirely new or is part of a bigger problem class. We conducted the literature review with this in mind. We first focus on the three studies mentioned earlier, then we look at problem classes that have some similarity with dispatcher operations, and finally we review the literature on workload balancing measures. In our review, we found that flight dispatcher scheduling presents a new problem and its uniqueness stems from the multi-tasking and dynamic nature of the workload.

The assignment of flight tasks to dispatchers is studied in Agnetis and Smriglio (2000), where each flight has only one task associated to it and a dispatcher is assigned one task at a time. Each task requires a known amount of time to process and a due date. Under these assumptions, the problem resembles a parallel machine scheduling problem with due dates where the goal is to find a feasible assignment. The paper further assumes that some flights are already assigned to dispatchers, which corresponds to unavailable machine times in the machine scheduling context, and that the task processing times take a given set of choices. The authors use a constraint satisfaction approach and explore solution properties as a function of processing times. Santos

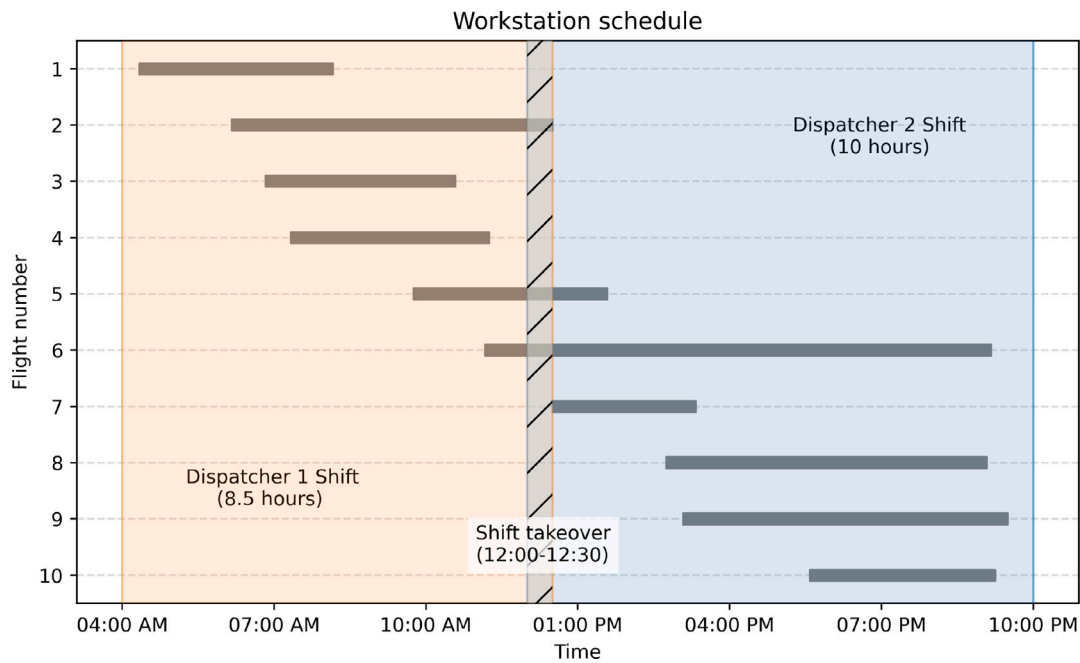


Fig. 1. Illustration of an example workstation schedule with 2 dispatcher shifts.

et al. (2011) study the assignment of flights to workstations under several goals and requirements. They attempt to capture the load requirement of a flight by dividing the workday into hourly intervals and defining the task requirement per hour as the time that needs to be allocated to the flight during that hour, e.g., a flight from 13:30 to 14:15 may have a 10-min task in hour 13, and a 5-min task in hour 14. The authors make an additional simplifying assumption that a flight may be assigned to different desks during different hours. The goals to be optimized include balancing workload which is measured as the hourly processing time assigned to a dispatcher, minimizing dispatcher over-utilization (i.e., being assigned more than 60 min of processing time), and minimizing the assignment of same flight tasks to different workstations. Rayner (2021) considers a similar problem to the one studied in this paper but focuses on the objective of minimizing the sum of peak workloads over all workstations. A Lagrangian relaxation and column generation were devised for the problem, along with construction and improvement heuristics. The construction heuristic assigns flights, one by one, to workstations with minimum workload at departure times. The improvement heuristic swaps flight assignments between workstations. Numerical testing is carried for flights of a European Airline and American Airlines, containing 50–434 flights and for 15, 20, and 25 flights per workstation.

These three studies provide valuable insights into dispatcher scheduling within the airline industry but fall short of modeling the realistic problem. Agnetis and Smriglio (2000) assume that a dispatcher deals with only one flight at a time, which is not true in practice. A dispatcher usually manages multiple flights, some in planning and others en-route, at any given time. Ignoring the multi-tasking feature of a dispatcher's work leads to inaccurate assessment of their load. Furthermore, Agnetis and Smriglio (2000) do not take workload minimization into account as they aim to find a feasible solution. Santos et al. (2011) models dispatcher workload but resort to representing the task load requirement as time per hour, which fails to account for due dates and consequently, does not represent an accurate model. For example, a 10-min task and two 5-min tasks represent an equivalent workload, but if both 5-min tasks must be finished earlier than the 10-min task, say at the 7th and 10th minutes of the hour, not only they would represent a higher workload for their duration but also the dispatcher may not have enough time to finish them by the set deadline.

Moreover, they split flight tasks between workstations, leading to flights being assigned to multiple dispatchers, which is not the case in practice. Rayner (2021) focuses on workload balancing but considers the workload only at flight departure times. Moreover, minimizing the sum of peak workloads of workstations might not necessarily achieve a balance, and makes the problem trivial. A trivial solution is to assign all flights to a single workstation. This solution leads to the worst-case workload imbalance. Our problem setting and objectives are not trivial as we capture more realistically the complex, time-dependent nature of dispatcher workload.

Alongside flight dispatchers, air traffic control specialists (ATC) provide ground support for flights.

Since both ATC and flight dispatchers plan commercial air traffic for different flights, they share similar workload challenges related to multi-tasking and unforeseen changes. ATC are government employees responsible for ensuring safety and order in air traffic. They manage flight operations at the airport and ensure that flight paths are followed and that there is sufficient separation. If necessary, they can make changes to the aircraft speed and altitude to manage the traffic (Ibáñez Gijón et al., 2023). A study on the air traffic controllers' mental load by Alharasees and Kale (2024) finds that the communication and workload are the most crucial factors affecting the operator's efficiency. The study relates the operator workload to four characteristics: human behavior, skills, tacit knowledge, and practice. There is a vast literature on ATC operations, studies include rostering (Pavlović et al., 2023), capacity planning (Starita et al., 2020), modeling and predicting the workload (Ibáñez Gijón et al., 2023; Alharasees and Kale, 2024), and traffic scheduling (Ikli et al., 2021; Mannino et al., 2021). The nature of the work makes the critical questions for dispatchers and ATCs inherently distinct. While workload balance is a common goal, the mechanisms to achieve it are different. ATC workload is determined by the preset airspace sector and consequently the flights that use it. On the other hand, the workload of the flight dispatcher may be managed by optimizing the assignment of flights to workstations. Unlike the flight dispatcher workload, the ATC workload is impacted by the airport and airspace they manage, which are not done for specific flights as in flight dispatchers, but are assigned ahead of time based on airspace. There is no literature, however, that tackles dynamic and time-varying load optimization based on airspace assigned, that would apply to flight dispatchers.

Dispatcher scheduling may resemble known problems already studied in the literature that consider workload balance such as **parallel machine scheduling** (Schwerdfeger and Walter, 2018). Parallel machine scheduling aims to assign jobs with known processing times to parallel machines. The load, as measured by the processing times, does not vary over time and the completion time is the objective often minimized (Ying et al., 2024).

Another closely related problem setting is that of service personnel scheduling where multiple tasks are simultaneously handled by single personnel within a work shift. Nurses and doctors, for example, may encounter a similar workload structure as flight dispatchers (Legrain et al., 2015; Schoenfelder et al., 2020). Nurses are responsible for providing care to patients and monitoring their status, they have specialized training that designates the unit they can work in. Considering that they monitor multiple patients simultaneously, face unpredictable spikes in workload as emergencies arise, and are assigned to specific wards, their workload resembles that of flight dispatchers. The literature, however, does not model the dynamic nature of the workload over time. Instead, the cumulative workload is used. Patient-to-nurse ratio is a measure used for modeling the nurse workload, and the patient-to-nurse assignments need to have this ratio to be as close as possible to a target value, which may be set by the government or hospital depending on the location. Anh (2022), for example, study the fair allocation of service tasks to healthcare workers by constraining the disparity in cumulative time it takes workers to finish their assigned tasks. The workload measure is reduced to the cumulative duration of all assigned tasks and does not explicitly model complexity nor load variations over time. In nurse scheduling the purpose is to assign nurses to shifts while accounting for nurse preferences, qualifications, work rules, etc. This problem has no similarity to the dispatcher workload balancing problem. For example, the study by Schoenfelder et al. (2020) addresses the scheduling problem while also considering patient transfers between units. Other studies that considered static workload include (Maenhout and Vanhoucke, 2010; Burke et al., 2010).

These static models consider balancing the total amount of work assigned to nurses, as opposed to modeling a time-dependent workload and balancing it over time, which is the case in flight dispatchers. Moreover, nurse scheduling consists of various restrictions and features, such as patient transfers, unit and specialty restrictions, and assignment quotas, whereas the assignments of flight dispatchers do not have such constraints. As a result, these methods are unsuitable in the context of flight dispatchers.

The workload balancing problem of nurses and personnel scheduling is different than that of flight dispatchers in terms of how the workload is defined. The workload in the former is one dimensional and represents the total load over time, which aims to balance the cumulative load as opposed to over time. Flight dispatchers encounter varying levels of workload over time. As flights depart and land, they require different tasks and loads from the dispatcher based on their status.

To measure workload balance, the literature has suggested multiple metrics depending on the problem studied. The work of Stecké (1983) on flexible manufacturing systems forms the groundwork of workload balancing measures. They focus on balancing various aspects of the manufacturing process, such as the processing times of machines, the number of machines used to process each part, and tool magazine utilization rates. To measure the workload balance, they utilize functions such as (i) maximum workload, (ii) maximum pairwise difference in workload, and (iii) sum of squared pairwise differences in workload. They formulate these problems as mixed integer programming problems and linearize the measure (iii) using cross-product terms. Kumar and Shanker (2001) provide a detailed comparison of the effectiveness of different workload balancing objectives. Their results indicate that maximizing machine utilization balances the workload and makes most objectives ineffective. They also find that minimizing the maximum

workload and the pairwise difference are the least and most effective measures for balancing workload, respectively.

However, it should be noted that these results are for the flexible manufacturing systems where the characteristics of a job are different from those of a flight in flight dispatcher scheduling. The difference is in the availability of jobs. In flexible manufacturing systems, a job may be processed at any time once the prerequisite processes are completed, which are also determined by the schedule. In flight dispatching, a flight has specific intervals of time during which the dispatcher must complete the related tasks. Vanheusden et al. (2022) address the workload balancing problem in warehouse order-picking operations and extend such metrics to multiple sets as: the sum of the peak workload in each zone, the sum of variance in the workload of each zone over time, the sum of absolute deviations from the median workload in each zone over time, and the lexicographic peak workload in zones.

The literature review revealed that current work on flight assignment and dispatcher scheduling makes limiting assumptions on load measures, on flight-to-dispatcher assignment, and on workload balancing. In this work, we model a more realistic setting that accounts for multi-tasking and introduces a load definition that is not time-based. This turns out to be, to the best of our knowledge, a new problem that has not been studied in literature. In the following section, we formally define the problem under study, we detail the load definition that we adopt, and we present a nonlinear mathematical model along with a linearization.

3. The dispatcher workload balancing problem

The flight dispatcher workload balancing problems attempts to find an assignment of flights to workstations such that the workload is balanced among workstations. Each flight is assigned to exactly one workstation, and it is managed by the dispatcher working at the workstation at the time. Flights have scheduled departure and arrival times that dictate the time-varying load required by the assigned dispatcher. The load represents the cognitive and physical load that a flight requires from the dispatcher, which may depend on the flight path, its time, aircraft characteristics, and airports. The load varies over time as the flight changes status, and as a result, the workload of a dispatcher changes as the load requirements of their assigned flights changes. It should be noted that the terms dispatcher and workstation are used interchangeably in this context, as a workstation has one dispatcher always working at it.

To illustrate these aspects on an example, let us consider 3 flights and 2 workstations. The flights depart at 6:08 am, 6:52 am, and 10:08 am, and land at 4:58 pm, 5:02 pm, and 8:03 pm, respectively. Planning for a flight starts 150 min before departure and lasts until 30 min before departure. Following the planning tasks, there is no task or load required from the dispatcher until the flight departs, after which the en-route tasks begin and last until the flight arrives at the destination airport. Fig. 2 illustrates the distribution of load requirements of the three flights over time. Let us consider two possible flight-workstation assignments where flights are assigned to workstations (1, 1, 2) and (1, 2, 2), respectively. Under a workload model where workload of a workstation at any given time is equal to the total load requirement of its assigned flights at the given time, the workload distribution of the two solutions would vary over time as depicted in Fig. 3.

Although the question of whether solution 1 or 2 has a better-balanced workload depends on the balance measure, solution 2 is more likely to be preferred by most measures because of its smaller difference in workstation workloads over time.

In order to calculate workstation workload, a set of time-points is required. While defining this set in equal time intervals such as minutes or seconds may be accurate to capture workload variation over time, it is neither necessary nor efficient. It is not necessary since the variation in load over time is due to flights changing status, so it is sufficient

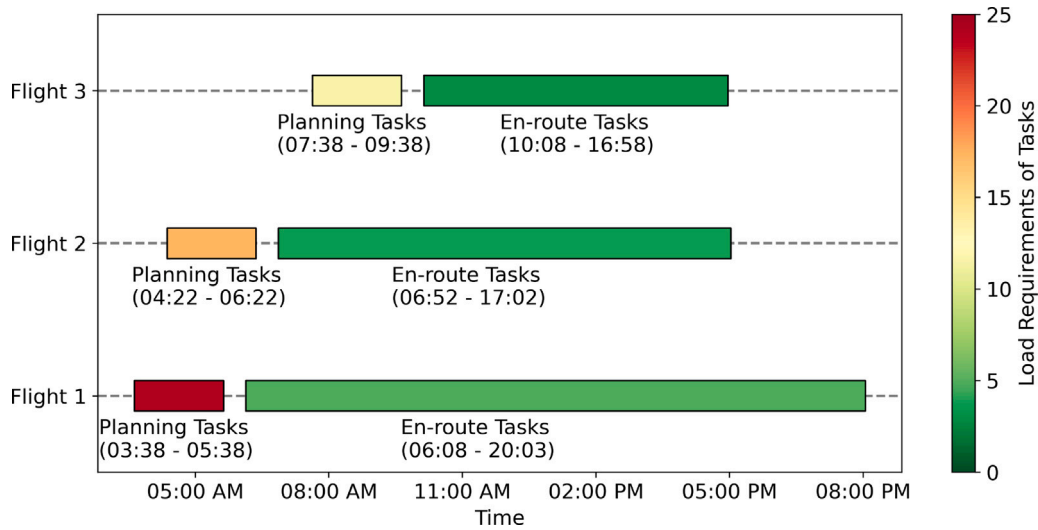


Fig. 2. Variation of required load for three flights, departing at 6:08 am, 6:52 am, 10:08 am, respectively.

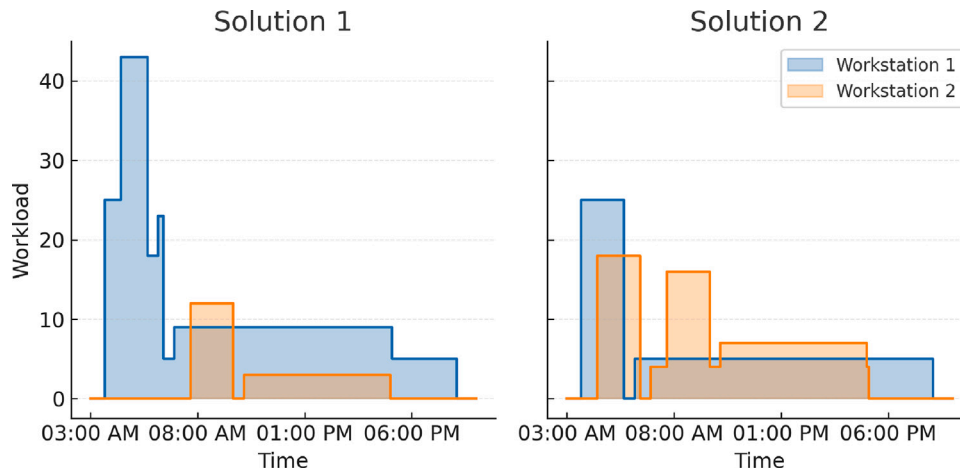


Fig. 3. Comparison of workload in two different assignments where workstation assignments of flights are (1, 1, 2) and (1, 2, 2), respectively.

to only consider the set of time-points where at least one flight has a change in load requirement. Similarly, it is not efficient because if two time-points are identical in terms of flight load requirements, they will also have an identical workload, and the number of time-points can be reduced. Let us denote by T the set of time-points at which there is a change in the load requirement of any flight. For the example instance provided in Fig. 2, T is {3:38, 4:22, 5:38, 6:08, 6:22, 6:52, 7:38, 9:38, 10:08 am; 4:58, 5:02, 8:03 pm}, i.e., whenever planning or en-route tasks of a flight start or end. Note that the cardinality of T is not necessarily a multiple of the number of flights since some time-points may be common to multiple flights and the elements of T are all distinct.

The distribution of flight load and workstation workload gets more complex as the number of flights increases. Fig. 4 illustrates an example instance that consists of 50 flights and 2 workstations. Ideally, the load would be distributed equally among workstations resulting in equal workload for all. This is unlikely to happen in practice given the discrete and the time-varying nature of load and the fact that a flight load cannot be split between workstations. Hence, we aim to balance workload under two measures: minimizing the peak workload and minimizing total absolute deviation from the average workload over time. Formally, given a set of flights I , workstations K , discrete time-points T , and flight load requirement h_{it} of flight $i \in I$ at time $t \in T$, the Flight Dispatcher Peak Workload Balancing Problem (PWP)

aims to find an assignment of flights to workstations that minimizes the maximum workload observed over $t \in T$ and $k \in K$. On the other hand, the Flight Dispatcher Absolute Workload Deviation Balancing Problem (ADP) aims to find an assignment that minimizes the total absolute deviation of workstation workloads from the average workload observed over $t \in T$.

The load of flight $i \in I$ at time $t \in T$, h_{it} , is unitless and reflects the relative load level among flights and tasks, e.g., if the planning task requires twice as much as the en-route task, the load associated with planning and en-route must reflect that same ratio. It is flight dependent as load may vary between flights, e.g., regional, international, as well as during different phases of the flight, such as planning and en-route. It is commonly said that a dispatcher has about 5 flights en-route for every flight they are planning at a given time (Collier and Bartels, 2021). The load may also depend on the dispatcher given their experience, which we do not consider in this research as we focus on the assignment of flights to workstations and not on dispatcher scheduling. The latter is in and of itself an important research question that needs to consider personnel scheduling aspects like seniority, preference, and availability. In calculating workstation workload using as the sum of assigned flight loads, we assume that flights do not share common characteristics that might affect their combined workload, which might not be the case in the real world. For example, when planning for two flights that have the same origin

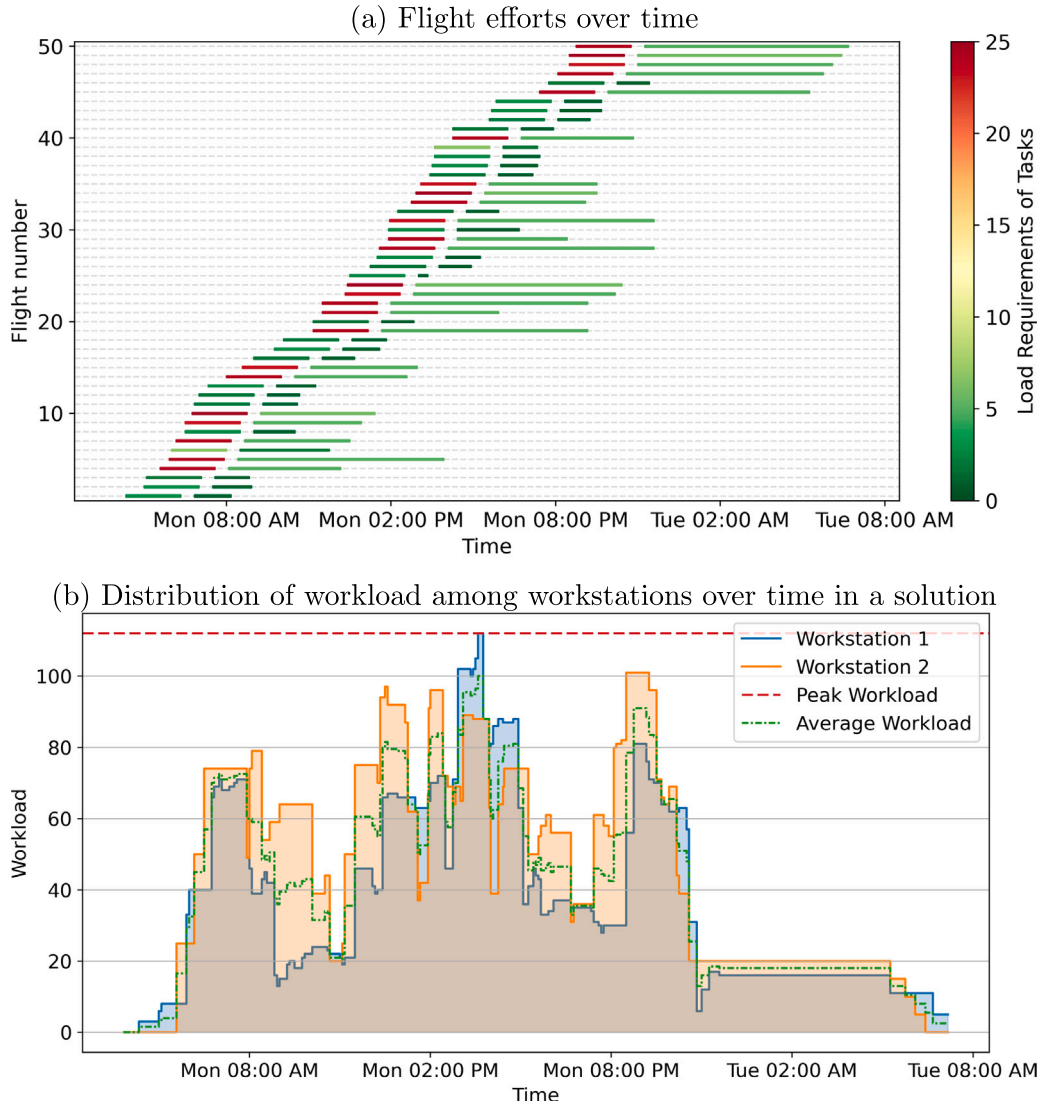


Fig. 4. Flight load requirements and distribution of workload in a solution for an instance with 50 flights and 2 workstations.

airport or that fly similar routes, the dispatcher may face a reduced load due to the similarity between the tasks. Incorporating workload functions that account for learning effects, dispatcher experience, and other dependencies requires rigorous research that is lacking in the current literature.

We denote the assignment of flight i to workstation k by a binary decision variable x_{ik} , which takes value 1 if the flight is assigned to the workstation and 0 otherwise. We use $f(\mathbf{x}, \mathbf{h})$ to represent a general function that measures the workload imbalance in a given assignment $\mathbf{x}$ based on the flight load requirements $\mathbf{h}$. The workload imbalance measure f produces a scalar value, which is minimized by the assignment with optimal workload balance. Given this notation, general mathematical model for the Flight Dispatcher Workload Balancing Problem is:

$$\min f(\mathbf{x}, \mathbf{h}) \quad (3.1)$$

$$\text{s.t. } \sum_{k \in K} x_{ik} = 1 \quad \forall i \in I \quad (3.2)$$

$$x_{ik} \in \{0, 1\} \quad \forall i \in I, k \in K. \quad (3.3)$$

The objective function (3.1) minimizes the workload balance measure function f , constraints (3.2) assign each flight to exactly one

workstation, and constraints (3.3) set the domains of assignment variables to binary. The measure function f is the key factor affecting the difficulty of the problem, as it may comprise nonlinear terms.

When minimizing the peak workload,

$$f(\mathbf{x}, \mathbf{h}) = \max_{k \in K, t \in T} \sum_{i \in I} h_{it} x_{ik} \quad (3.4)$$

which is a nonlinear function and may be linearized by introducing decision variable $F^{\max}$ to represent the maximum workload observed across all workstations and over all time-points. The Flight Dispatcher Peak Workload Balancing Problem (PWP) is:

$$(PWP) : \min F^{\max} \quad (3.5)$$

$$\text{s.t. } \sum_{k \in K} x_{ik} = 1 \quad \forall i \in I \quad (3.6)$$

$$\sum_{i \in I} h_{it} x_{ik} \leq F^{\max} \quad \forall t \in T, k \in K \quad (3.7)$$

$$x_{ik} \in \{0, 1\}, F^{\max} \geq 0 \quad \forall i \in I, k \in K. \quad (3.8)$$

The objective function (3.5) minimizes the peak workload $F^{\max}$, and constraints (3.7) set the peak workload $F^{\max}$ as the maximum over $t \in T$ and $k \in K$.

For the absolute deviation workload, let $\eta_t = \frac{\sum_{i \in I} h_{it}}{|K|}$ denote the average workload per workstation at time $t \in T$. The balance measure function takes the form:

$$f(\mathbf{x}, \mathbf{h}) = \sum_{k \in K} \sum_{t \in T} \left| \sum_{i \in I} h_{it} x_{ik} - \eta_t \right|, \quad (3.9)$$

which minimizes the total absolute deviation of workstation workloads from the average workload η_t over $t \in T$. The measure is nonlinear due to the absolute value function, which may be linearized introducing two decision variables F_{ik}^+ and F_{ik}^- that represent the workload of workstation $k \in K$ above and below the average workload η_t at $t \in T$. The Flight Dispatcher Absolute Workload Deviation Balancing Problem (ADP) is:

$$(ADP) : \min \sum_{k \in K} \sum_{t \in T} F_{ik}^+ + F_{ik}^- \quad (3.10)$$

$$\text{s.t.} \quad \sum_{k \in K} x_{ik} = 1 \quad \forall i \in I \quad (3.11)$$

$$\sum_{i \in I} h_{it} x_{ik} = \eta_t + F_{ik}^+ - F_{ik}^- \quad \forall k \in K, t \in T \quad (3.12)$$

$$x_{ik} \in \{0, 1\}, F_{ik}^+, F_{ik}^- \geq 0 \quad \forall i \in I, k \in K, t \in T. \quad (3.13)$$

The objective function (3.10) minimizes the total workload deviations over workstations and time-points. Constraints (3.12) set F_{ik}^+ and F_{ik}^- as the positive and negative parts of the difference between workload $k \in K$ workload at time $t \in T$ and the average workload at time $t \in T$. Constraints (3.13) set the domains of the variables.

While in this work, we model peak workload and absolute deviation as balance measures, model (3.1–3.3) is general and may apply to any workload balance measure. Though, solution of the model for specific measures will depend on that measure.

Even though we can linearize the nonlinear objectives for both (PWP) and (ADP), the resulting mixed integer programming problems are challenging to solve. As reported in Section 6.2, existing solvers fail to solve real world size instances within hours of computational time. The main challenges are the inherent nonlinear objectives and large scale. For example, there may be thousands of flights in each day, which could amount to hundreds of thousands of decision variables. We thus focus on developing solution methods that can solve real world instances quickly. In the following section, we propose a Lagrangian relaxation of (PWP) and derive closed form expressions of the Lagrangian bound.

4. Lagrangian relaxation and lower bounding

In this section, we explore a Lagrangian relaxation (Geoffrion, 1974) on the Flight Dispatcher Peak Workload Balancing Problem, examine characteristics of the resulting subproblem and prove that the Lagrangian bound has a closed-form expression.

In an effort to solve the Flight Dispatcher Peak Workload Balancing Problem (PWP) efficiently, we propose to relax assignment constraints (3.6) with unrestricted Lagrangian multiplier μ_i , $i \in I$. The resulting Lagrangian subproblem is:

$$(SP) : z_{SP} = \min F^{\max} - \sum_{i \in I} \sum_{k \in K} \mu_i x_{ik} \\ \text{s.t.} \quad \sum_{i \in I} h_{it} x_{ik} \leq F^{\max} \quad \forall t \in T, k \in K \\ F^{\max} \geq 0, x_{ik} \in \{0, 1\} \quad \forall i \in I, k \in K.$$

The Lagrangian lower bound $\mathcal{LR}_{MW}$ is the solution of the following Lagrangian dual problem:

$$(DP) : \max_{\mu} z_{SP} + \sum_{i \in I} \mu_i,$$

which may be reformulated as a linear program, leading to the Lagrangian master problem:

$$(MP) : z_{MP} = \max \quad \theta + \sum_{i \in I} \mu_i \\ \text{s.t.} \quad \theta + \sum_{i \in I, k \in K} x_{ik}^h \mu_i \leq F_h^{\max} \quad \forall h \in H \\ \theta, \mu_i \in \mathbb{R} \quad \forall i \in I.$$

where $h \in H$ represents the indices of solutions x_{ik}^h and $F_h^{\max}$ feasible to SP , and θ is an auxiliary variable used to linearize the master problem corresponding to the subproblem objective $z_{SP} = \min_{h \in H} F_h^{\max} - \sum_{i \in I, k \in K} x_{ik}^h \mu_i$. The dual of MP is the Dantzig–Wolfe master problem:

$$(DW) : \min \sum_{h \in H} \alpha_h F_h^{\max} \\ \text{s.t.} \quad \sum_{h \in H} \alpha_h = 1 \quad (\text{dual variables: } \theta) \\ \sum_{h \in H} \alpha_h x_{ik}^h = 1 \quad \forall i \in I, k \in K \quad (\text{dual variables: } \mu_i) \\ \alpha_h \geq 0 \quad \forall h \in H.$$

Different approaches exist to solve the Lagrangian dual problem. Two of the most common algorithms are subgradient optimization and cutting plane/column generation methods. However, in this particular case, the Lagrangian dual problem DP may be solved explicitly. In the remainder of this section, we explore the mathematical properties of the subproblem SP and the Lagrangian dual problem DP and provide a closed-form expression of its optimal solution.

4.1. Properties of the Lagrangian subproblem

To study the properties of the subproblem for its effective solution, we first investigate the relationship between SP and its linear relaxation, which we call LSP . In particular, we show that SP does not have the integrality property, and the maximum workload is not necessarily the same in the optimal solutions of LSP and SP . Second, we prove an important property of the subproblem that greatly helps with its solution.

For the integrality property, let us consider an example with 4 flights $I = \{1, 2, 3, 4\}$, 2 workstations $K = \{1, 2\}$, three-time-points $T = [1, 2, 3]$, Lagrangian multipliers $\mu_i = [12, 9, 12, 9]$, and load matrix:

$$h_{it} = \begin{bmatrix} 0 & 0 & 16 \\ 0 & 24 & 24 \\ 16 & 0 & 0 \\ 24 & 24 & 0 \end{bmatrix}$$

The optimal solution of SP is $x_{ik} = 1$ for all $i \in I, k \in K$ with objective -36 and $F^{\max} = 48$, whereas LSP has solution

$$x_{ik}^{LSP*} = \begin{bmatrix} 1 & 1 \\ 0.67 & 0.67 \\ 1 & 1 \\ 0.67 & 0.67 \end{bmatrix}$$

with objective -40 and $F^{\max} = 32$, different from that of SP . Therefore, the SP does not have the integrality property. Next, we show that the Lagrangian bound has a closed-form solution by first showing that the subproblem SP has an equivalent reduced problem in Proposition 1, we then use this result to derive the closed-form Lagrangian bound in Proposition 2.

Proposition 1. The Lagrangian subproblem SP :

1. has an optimal solution x^* in which all workstations are assigned the same set of flights. i.e., $x_{ik_1}^* = x_{ik_2}^* \forall k_1, k_2 \in K$.
2. reduces to an identical subproblem of this form:

$$(RSP) : \min F^{\max} - |K| \sum_{i \in I} \mu_i x_i$$

$$\begin{aligned} \text{s.t. } & \sum_{i \in I} h_{it} x_i \leq F^{\max} & \forall t \in T \\ & F^{\max} \geq 0, x_i \in \{0, 1\} & \forall i \in I, \end{aligned}$$

where $x_i, i \in I$, is a binary decision variable that takes value of 1 if flight is assigned and zero otherwise.

Proof. Let us write SP in the following equivalent form where $F_k^{\max}$ is the peak workload of workstation k :

$$\begin{aligned} \min \quad & F^{\max} - \sum_{i \in I, k \in K} \mu_i x_{ik} \\ \text{s.t. } & \sum_{i \in I} h_{it} x_{ik} \leq F_k^{\max} & \forall t \in T, k \in K \\ & F_k^{\max} \leq F^{\max} & k \in K \\ & F^{\max}, F_k^{\max} \geq 0, x_{ik} \in \{0, 1\} & \forall i \in I, k \in K. \end{aligned}$$

The proof seeks to find a lower bound and an upper bound, corresponding to a feasible solution, that coincide. For the lower bound, let us resort to the following Lagrangian relaxation where the second set of constraints are relaxed with Lagrange multipliers $\beta_k \geq 0, k \in K$, leading to

$$\begin{aligned} z_{SP}(\beta) = \min \quad & \left(1 - \sum_{k \in K} \beta_k\right) F^{\max} \\ & + \sum_{k \in K} \beta_k F_k^{\max} - \sum_{i \in I, k \in K} \mu_i x_{ik} \\ \text{s.t. } & \sum_{i \in I} h_{it} x_{ik} \leq F_k^{\max} & \forall t \in T, k \in K \\ & F^{\max}, F_k^{\max} \geq 0, x_{ik} \in \{0, 1\} & \forall i \in I, k \in K, \end{aligned}$$

where β is the vector with items $\beta_k, k \in K$. For any $\beta_k \geq 0$, a valid lower bound is $z_{SP}(\beta)$. We are particularly interested in $\beta_k^* = \frac{1}{|K|}$ and its corresponding subproblem:

$$\begin{aligned} z_{SP}(\beta^*) = \min \quad & \sum_{k \in K} \frac{1}{|K|} F_k^{\max} - \sum_{i \in I, k \in K} \mu_i x_{ik} \\ \text{s.t. } & \sum_{i \in I} h_{it} x_{ik} \leq F_k^{\max} & \forall t \in T, k \in K \\ & F_k^{\max} \geq 0, x_{ik} \in \{0, 1\} & \forall i \in I, k \in K. \end{aligned}$$

The latter is decomposable into $|K|$ identical subproblems:

$$\begin{aligned} (RSP) : \quad z_{RSP} = \min \quad & \frac{1}{|K|} F^{\max} - \sum_{i \in I} \mu_i x_i \\ \text{s.t. } & \sum_{i \in I} h_{it} x_i \leq F^{\max} & \forall t \in T \\ & F^{\max} \geq 0, x_i \in \{0, 1\} & \forall i \in I. \end{aligned}$$

Let us denote the optimal solution of (RSP) as $F_*^{\max}$ and $x_i^*, i \in I$. The lower bound is then

$$|K| \left(\frac{1}{|K|} F_*^{\max} - \sum_{i \in I} \mu_i x_i^* \right) = F_*^{\max} - |K| \sum_{i \in I} \mu_i x_i^*. \quad (4.1)$$

Now let us build a feasible solution to SP , which we recall as:

$$\begin{aligned} \min \quad & F^{\max} - \sum_{i \in I} \sum_{k \in K} \mu_i x_{ik} \\ \text{s.t. } & \sum_{i \in I} h_{it} x_{ik} \leq F^{\max} & \forall t \in T, k \in K \\ & F^{\max} \geq 0, x_{ik} \in \{0, 1\} & \forall i \in I, k \in K. \end{aligned}$$

Based on x_i^* , let $\bar{x}_{ik} = x_i^* k \in K$ and $\bar{F}^{\max} = F_*^{\max}$, which is obviously feasible to SP . Its corresponding objective gives the following an upper bound:

$$\bar{F}^{\max} - \sum_{i \in I} \sum_{k \in K} \mu_i \bar{x}_{ik} = F_*^{\max} - |K| \sum_{i \in I} \mu_i x_i^* \quad (4.2)$$

As the lower bound (4.1) is the same as the upper bound (4.2), $(\bar{x}_{ik}, \bar{F}^{\max})$ is optimal to SP .

In fact, for any optimal solution of subproblem SP , every pair of workstations $k_1, k_2 \in K$ has identical flight assignments with respect to maximum workload $\max_{t \in T} \sum_{i \in I} h_{it} x_{ik}$ and total Lagrangian multiplier weight $\sum_{i \in I} \mu_i x_{ik}$. In other words, $\max_{t \in T} \sum_{i \in I} h_{it} x_{ik_1} = \max_{t \in T} \sum_{i \in I} h_{it} x_{ik_2}$ and $\sum_{i \in I} \mu_i x_{ik_1} = \sum_{i \in I} \mu_i x_{ik_2}$ for $k_1, k_2 \in K$.

To show this, let $x^* = \{x_{ik} \in \{0, 1\} \forall i \in I, k \in K\}$ denote an optimal solution to the subproblem. The total penalty for workstation k , denoted by v_k , is given by $\sum_{i \in I} \mu_i x_{ik}^*$. Furthermore, we define $v^{\max}$ as the maximum penalty across all workstations: $v^{\max} = \max_{k \in K} v_k$. Consider two subsets of workstations: A and B such that $A, B \subseteq K$. We define A as the set of workstations for which $F_k^{\max}$ is maximized. Similarly, B consists of workstations where v_k is maximized. Hence,

$$F^{\max} = F_k^{\max}, \forall k \in A \text{ and } v^{\max} = v_k, \forall k \in B.$$

Let us establish that x^* is optimal if and only if sets A and B coincide. First, suppose $A \not\subseteq B$, meaning $A \setminus B \neq \emptyset$. For a workstation l in A but not in B , we have:

$$F_l^{\max} = F^{\max}, \quad v_l < v^{\max}.$$

For any workstation l' in B ,

$$v_l < v_{l'}, \quad F_l^{\max} \geq F_{l'}^{\max}.$$

Considering a modified solution where workstation l follows the assignments of workstation l' in x^* (with all other assignments remaining unchanged), the objective becomes:

$$\max_{k \in K \setminus \{l\}} F_k^{\max} - \left(\sum_{k \in K} v_k - v_l + v_{l'} \right).$$

This is strictly lower than the objective value of x^* , indicating that workstation l does not appear in x^* . This leads us to conclude $A \subseteq B$ for optimality. Second, for the reverse scenario, assume $B \not\subseteq A$ or $B \setminus A \neq \emptyset$. A workstation $l \in B \setminus A$ satisfies:

$$F_l^{\max} < F^{\max}, \quad v_l = v^{\max}.$$

If all workstations follow the assignments of l , the objective becomes:

$$F_l^{\max} - |K| v_l$$

which is strictly lower than x^* . Thus, for optimality, $B \subseteq A$. Therefore, sets A and B are identical in an optimal solution.

Next, we show that A and B collectively include all workstations K in optimal solutions. Suppose l' belongs to both A and B . For a workstation l not in A or B , we have:

$$F_l^{\max} < F^{\max}, \quad v_l < v^{\max}.$$

A solution where l adopts the assignments of l' yields a reduced objective value:

$$F^{\max} - \sum_{k \in K} v_k > F^{\max} - \left(\sum_{k \in K} v_k - v_l + v_{l'} \right).$$

Therefore, all workstations must be in both A and B for optimality, implying that all workstations have equal peak workloads and incur identical penalties in optimal solutions.

Proposition 1 implies that instead of solving SP which involves $|K| \parallel I|$ binary variables, we could just solve RSP with just $|I|$ binary variables. In that case, the Lagrangian master problem MP is reduced to:

$$\begin{aligned} (MP) : \quad z_{MP} = \max \quad & \theta + \sum_{i \in I} \mu_i \\ \text{s.t. } & \theta + |K| \sum_{i \in I} x_i^h \mu_i \leq F_h^{\max} & \forall h \in H \\ & \theta \in \mathbb{R}, \mu_i \in \mathbb{R} & \forall i \in I. \end{aligned} \quad (4.3)$$

In the following section, we provide a closed form expression for the optimal Lagrangian bound.

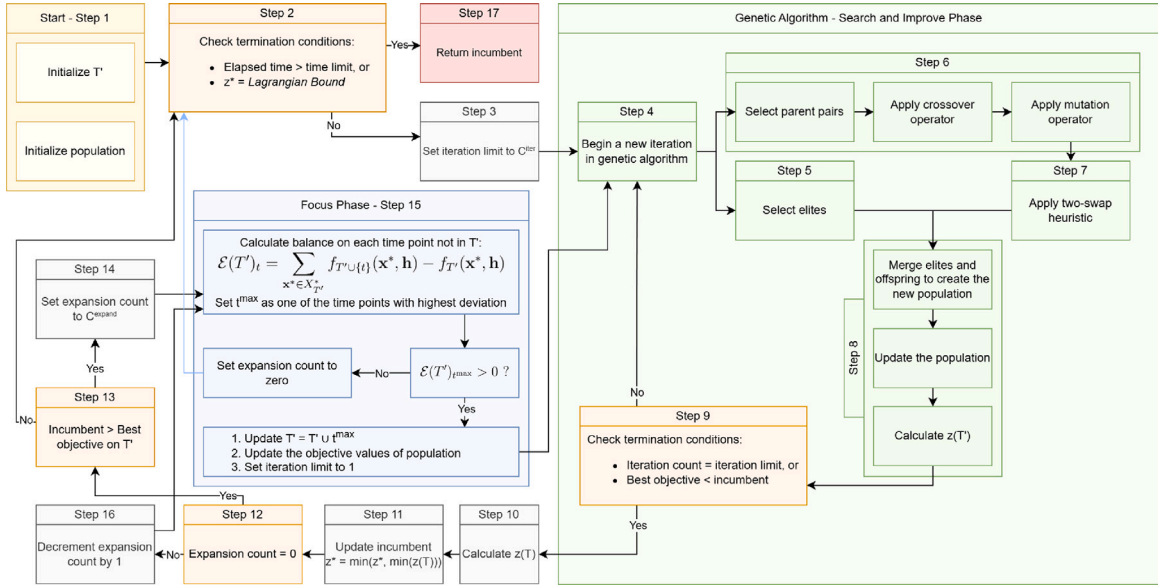


Fig. 5. Overview of FSI-GA.

4.2. The optimal Lagrangian bound

The reduced subproblem RSP provides the basis for deriving a closed form expression of the optimal Lagrangian lower bound. Proposition 2 shows that the master problem MP has a known optimal solution.

Proposition 2. The optimal objective value of the Lagrangian master problem MP is given by:

$$\max_{i \in T} \sum_{i \in I} \frac{h_{it}}{|K|}$$

Proof. First, we show that $\max_{i \in T} \sum_{i \in I} \frac{h_{it}}{|K|}$ is a lower bound for MP . Consider the dual solution given by $(\theta = 0, \mu_i = \frac{h_{is}}{|K|} \forall i \in I)$. Here, s corresponds to a time-point that maximizes the total load requirement of flights. Explicitly, s is:

$$s \in \operatorname{argmax}_{i \in T} \sum_{i \in I} h_{it}.$$

This solution $(\theta = 0, \mu_i = \frac{h_{is}}{|K|} \forall i \in I)$ satisfies constraints (4.3) since the left-hand side becomes equivalent to the workload at time s , which is always less than or equal to the peak workload $F_h^{\max}$:

$$(\theta = 0) + |K| \sum_{i \in I} x_i^h \left(\mu_i = \frac{h_{is}}{|K|} \right) = \sum_{i \in I} x_i^h h_{is} \leq F_h^{\max} = \max_{i \in T} \sum_{i \in I} h_{it} x_i^h.$$

Therefore, its objective, $\max_{i \in T} \sum_{i \in I} \frac{h_{it}}{|K|}$, is a lower bound. Next, we show that this value is also an upper bound for MP . Let us consider a relaxation of MP obtained by keeping constraints (4.3) for two feasible solutions of RSP . That is, we remove all constraints that correspond to feasible solutions other than these two. The first solution does not select any flights, $x_i = 0, i \in I$, whereas the second solution selects all flights: $x_i = 1, i \in I$. The corresponding relaxed master problem is:

$$\begin{aligned} \max \quad & \theta + \sum_{i \in I} \mu_i \\ \text{s.t.} \quad & \theta \leq 0 \end{aligned} \quad (4.4)$$

$$\theta + |K| \sum_{i \in I} \mu_i \leq \max_{i \in T} \sum_{i \in I} h_{it} \quad (4.5)$$

$$\theta \in \mathbb{R}, \mu_i \in \mathbb{R} \quad \forall i \in I.$$

Let us assume that its solution is $\bar{\theta}, \bar{\mu}_i i \in I$. Being a relaxation of MP , the corresponding objective $\bar{\theta} + \sum_{i \in I} \bar{\mu}_i$ is an upper bound of MP . Multiplying constraint (4.4) by $(|K| - 1)$ and adding the resulting inequality to constraint (4.5) leads to the following condition on the objective function:

$$|K|\bar{\theta} + |K| \sum_{i \in I} \bar{\mu}_i \leq \max_{i \in T} \sum_{i \in I} h_{it} \Rightarrow \bar{\theta} + \sum_{i \in I} \bar{\mu}_i \leq \max_{i \in T} \sum_{i \in I} \frac{h_{it}}{|K|},$$

which implies that $\max_{i \in T} \sum_{i \in I} \frac{h_{it}}{|K|}$ is an upper bound. Therefore, as this value is also previously shown to be a lower bound, the Lagrangian dual problem MP has an optimal objective value of $\max_{i \in T} \sum_{i \in I} \frac{h_{it}}{|K|}$. For the problem instances where h_{it} is integer valued and x_{ik} are binary, the optimal solution is integer valued, and the Lagrangian bound can be rounded up to $\lceil \max_{i \in T} \sum_{i \in I} \frac{h_{it}}{|K|} \rceil$.

As a result, the optimal Lagrangian lower bound is calculated directly without the need to solve the Lagrangian dual iteratively. In the following section, we devise a heuristic that exploits the Lagrangian bound.

5. A Lagrangian guided focus-search-improve heuristic

In practice, flight dispatcher operations often involve managing over thousands of flights per day. This leads to large-scale instances of the load balancing problem that need to be solved repeatedly and in short computational times. As a result, we develop a multi-layered genetic algorithm, which we refer as Focus-Search-Improve Genetic Algorithm (FSI-GA), that produces high quality solutions in very short times as shown in the numerical testing. The heuristic searches for solutions in three main phases: focus, search, and improve, which form a hierarchy in terms of how they affect the search for feasible solutions and the frequency in which they are applied. First, FSI-GA focuses the search by restricting the search space by considering only a subset of time-points $T' \in T$, thus reducing the computational cost of building and evaluating solutions. Second, a genetic algorithm first generates local solutions on the restricted space, then updates the pool of overall solutions by evaluating the workload balance over all time-points T . Third, solutions are further improved using a two-swap improvement heuristic. All three search phases are applied repeatedly by selectively augmenting T' and consequently expanding the search space. Genetic algorithm was chosen as the core solution exploration method mainly because almost all of its procedures can be parallelized to take advantage of multi-threading, which allows utilizing more of

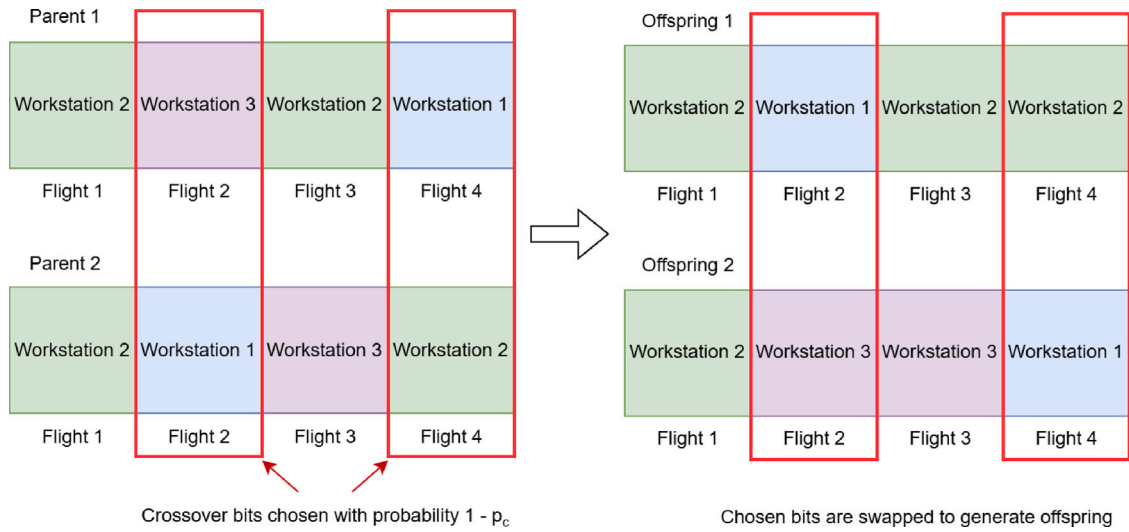


Fig. 6. An example of uniform crossover operation on two solutions for 5 flights and 3 workstations.

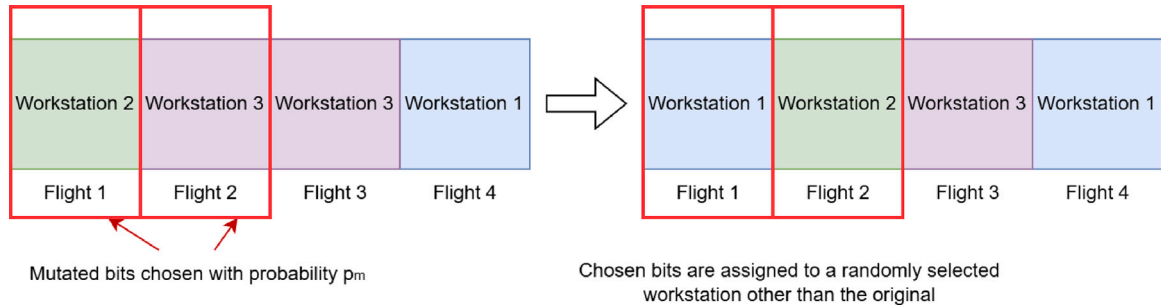


Fig. 7. An example mutation operation on a solution for 5 flights and 3 workstations.

the available computational resources. Fig. 5 gives an overview of the algorithm, and the three phases are explained next.

5.1. The search phase

The search phase is based on a Genetic Algorithm that uses conventional settings and employs crossover-mutation and elitism in generating new solutions and terminates when the incumbent equals the Lagrangian lower bound, or by time limit. Solutions are represented as integer vectors of size $|I|$ with values $\{1, 2, \dots, |K|\}$, indicating the workstation assignments of flights, e.g., if index i takes value k , flight i is assigned to workstation k in the generated solution. Example representations can be seen in Figs. 6 and 7.

To initialize the genetic algorithm, a population of solutions is initialized randomly by generating vectors of size $|I|$ with each element taking a random value from $\{1, 2, \dots, |K|\}$, i.e., each solution is a random permutation of $\{1, 2, \dots, |K|\}$.

To generate a new population, the C^{elite} solutions with best objective values are selected as elites and are added to the new population without modification. In addition, a select set of parent pairs are recombined and mutated. Selection is based on an exponential rank selector that assigns a selection weight $w_r = \frac{c^r(c-1)}{c^P-1}$ to a solution with rank r , where the best solution has rank $r = 1$, c is the rate parameter, and P is the population size. Each pair of solutions undergoes crossover and mutation operations to generate two new solutions that are added to the population. The crossover operator uses a uniform crossover where each bit of an offspring is treated separately and chosen from

either parent. Each of the two offspring has a designated parent from which they are more likely to inherit genes, this is controlled by the crossover probability p_c where the offspring inherits the gene from their designated parent with probability $1 - p_c$ or from the other parent with probability p_c . An example crossover operation is illustrated in Fig. 6. The mutation operator changes the bit to a different and randomly chosen value, which corresponds to changing the workstation assignment of a flight in the solution, Fig. 7 illustrates an example mutation.

5.2. The improvement phase

The Two-Swap Improvement Heuristic modifies a given solution by considering swapping workstation assignments of two flights. The first flight is selected so that, if assigned to a different workstation, the balance objective may improve. For ADP, this could be any flight, whereas for PWP, it corresponds to flights that are assigned to workstations with highest peak workload and active at a time-point where the workstation workload is at the peak, i.e., from set $I^1 = \{i_1 \in I | \exists t \in T, k \in K : x_{i_1 k} = 1, h_{i_1 t} > 0, k \in \arg\max_{j \in K} \sum_{j \in I} h_{j t} x_{j k}\}$. The second flight is selected such that, when swapped with the first flight, there is an improvement in the balance objective for their workstations over the times where either flight is active. For ADP, it means a decrease in the objective function value, while for PWP it means that the peak workload of the related workstations at times where either flight is active is less than the previous overall peak workload. Let $F_k = \max_{t \in T} \sum_{j \in I} h_{j t} x_{j k}$ denote the peak workload of workstation $k \in K$. For a given flight $i_1 \in I^1$, the set of eligible flights $I^2(i_1)$ can be characterized in two ways. For the PWP case, $i_2 \in I^2(i_1)$ whenever the following conditions hold:

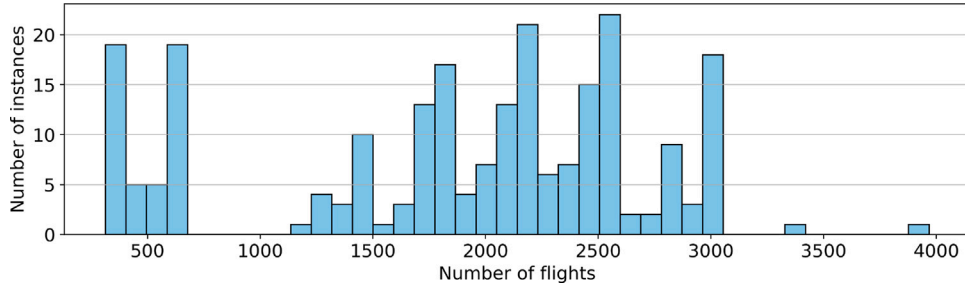


Fig. 8. Distribution of the number of flights in test instances.

Table 1

Flight summary statistics of the test instances, categorized by carriers.

Carrier	Instance count	Flights per instance			Time-points per instance		
		Min.	Avg.	Max.	Min.	Avg.	Max.
Alaska Airlines	24	527	611	640	325	355	390
American Airlines	24	1889	2101	2208	1331	1389	1448
Delta Air Lines	83	1706	2276	2627	1151	1262	1330
Endeavor Air	24	313	378	431	591	651	731
SkyWest Airlines	22	1415	1774	2363	1091	1166	1249
Southwest Airlines	35	2656	2965	3968	278	299	335
United Air Lines	19	1161	1416	1870	1107	1176	1334
Overall	231	313	1873	3968	278	955	1448

1. $x_{i_1 k_1} = 1$ and $x_{i_2 k_2} = 1$,
2. $\sum_{j \in I, j \neq i_1} h_{jt} x_{jk_1} + h_{i_2 t} < F_{k_1}$,
3. $\sum_{j \in I, j \neq i_2} h_{jt} x_{jk_2} + h_{i_1 t} < F_{k_2}$,
4. for all $t \in T$ with $h_{i_1 t} + h_{i_2 t} > 0$, there exist $k_1, k_2 \in K$.

For the ADP case, $i_2 \in I^2(i_1)$ whenever:

1. $x_{i_1 k_1} = 1$ and $x_{i_2 k_2} = 1$,
2. $\sum_{j \in I, j \neq i_1} h_{jt} x_{jk_2} + h_{i_2 t} - \eta_t < \sum_{t \in T} (F_{k_1 t}^+ + F_{k_1 t}^-)$,
3. $\sum_{j \in I, j \neq i_2} h_{jt} x_{jk_2} + h_{i_1 t} - \eta_t < \sum_{t \in T} (F_{k_2 t}^+ + F_{k_2 t}^-)$,
4. there exist $k_1, k_2 \in K$.

The improvement heuristic then searches the set $I^2(i_1)$ for $i_1 \in I^1$, and a swap is performed when an eligible flight $i_2 \in I^2(i_1)$ is found for $i_1 \in I^1$. The operation is applied recursively until no eligible swap can be found. For PWP, it should be noted that a swap does not necessarily decrease the global peak workload, but it does guarantee a decrease in peak workload of the assigned workstations of flights i_1 and i_2 at times where either flight is active.

5.3. The focus phase

The focus procedure initializes the subset T' by ranking time-points in non-increasing order of total flight load requirements, calculated as $\sum_{i \in I} h_{it}$ for $t \in T$, and selecting the first τ time-points. This may not yield entirely accurate solutions since the objective does not account for the workload balance outside T' . To address this, the algorithm regularly evaluates the solution objectives for two different time sets within its solution pool, incrementally expanding T' with the time-points that have the largest impact on the objective. Meaning that the added time-point is the one that has the largest peak workload exceeding that of T' for PWP, or the one that has the largest total absolute deviation from the average workload for ADP.

The incremental extension of T' serves two purposes: it refines the solution accuracy as the algorithm progresses and avoids sets of time-points that have similar workload. Workload similarity in this context refers to time-points in proximity (e.g., 15:01 and 15:02) where flights are at similar operational phases, requiring similar levels of dispatcher workload. Including time-points with similar workloads in

T' may result in redundant computations. The following error metric is introduced to update the set T' :

$$\mathcal{E}(T')_t = \sum_{\mathbf{x}^* \in X_{T'}^*} f_{T' \cup \{t\}}(\mathbf{x}^*, \mathbf{h}) - f_{T'}(\mathbf{x}^*, \mathbf{h}) \quad \forall t \in T, T' \subseteq T, \quad (5.1)$$

where $X_{T'}^*$ denotes the set of solutions on T' , and $f_T(\mathbf{x}, \mathbf{h})$ is the balance objective value for solution $\mathbf{x}$ and flight loads $\mathbf{h}$ calculated over set T . For a given time-point $t \notin T'$, the metric calculates the error in using T' to approximate the workload balance relative to $T' \cup \{t\}$. In particular, for a given set of best solutions in the population X and their corresponding best objective values, the procedure selects one of the time-point that maximizes the expressions (5.2) and (5.3) for PWP and ADP, respectively:

$$\operatorname{argmax}_{t \in T \setminus T'} \max_{\mathbf{x} \in X} \{0, \max_{k \in K} \sum_{i \in I} h_{it} x_{ik} - z^*\} \quad (5.2)$$

$$\operatorname{argmax}_{t \in T \setminus T'} \sum_{\mathbf{x} \in X} \sum_{k \in K} \left| \sum_{i \in I} h_{it} x_{ik} - \eta_t \right|. \quad (5.3)$$

Details are provided in Algorithm A.2.

6. Numerical testing

In this section, we report on extensive numerical testing performed on (PWP) and (ADP) to assess the quality of the Lagrangian lower bound calculated using the closed form expression, and the performance of the FSI-GA heuristic relative to the Lagrangian bound and to the commercial solver Cplex. We also analyze the effectiveness of the minimum peak workload objective function in achieving the main goal of balancing workload among workstations and over time. All testing is performed on real data instances that we describe in Section 6.1.

All computations are made on a computer that has an Intel i7-4790 processor with a speed of 3.6 GHz and 8 GB of RAM. Mathematical formulations are solved using Cplex 20.1 implemented in Java 18.0.1. Node files, produced when solving the mixed integer program as the branch-and-bound tree is explored, are stored in the memory with a 3 GB limit. Additional files that exceed the limit are stored on a hard-disk drive. The heuristic methodology is implemented in the same Java version utilizes a multi-threading structure limited to 8 threads, same as the Cplex solver.

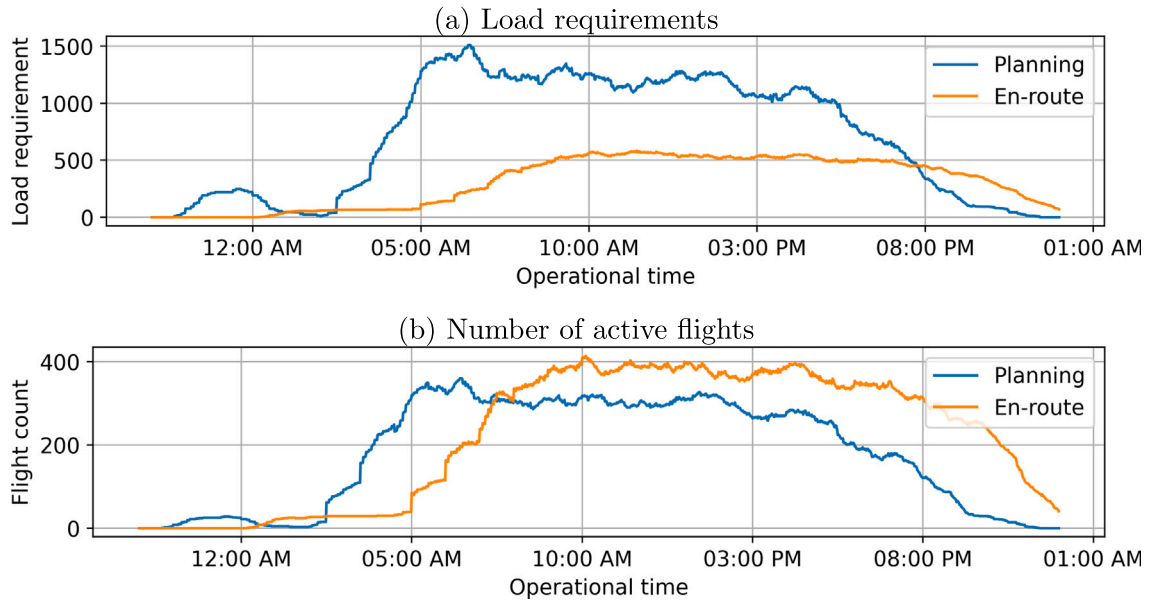


Fig. 9. Planning and en-route phases over the operational hours in the problem instance of Delta Air Lines 4 Sep 2019.

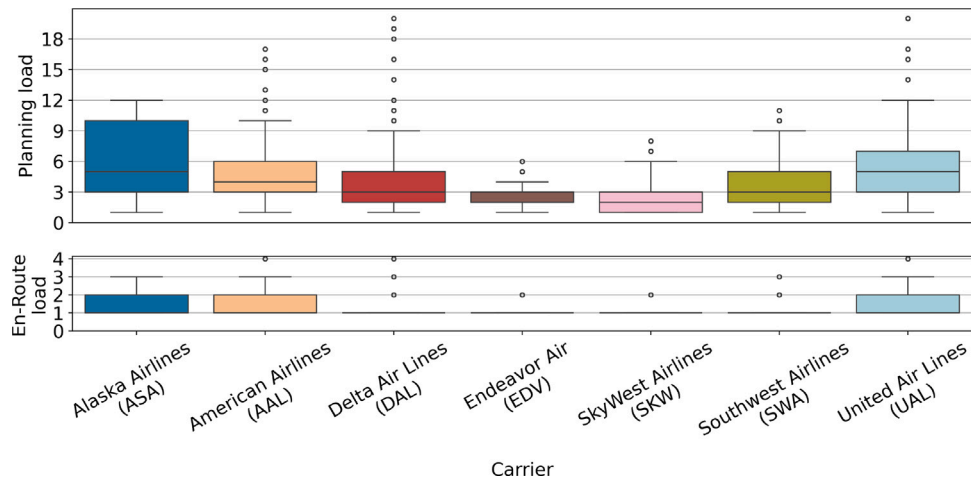


Fig. 10. Distribution of planning and en-route load requirements of flights in the generated instances, categorized by carrier.

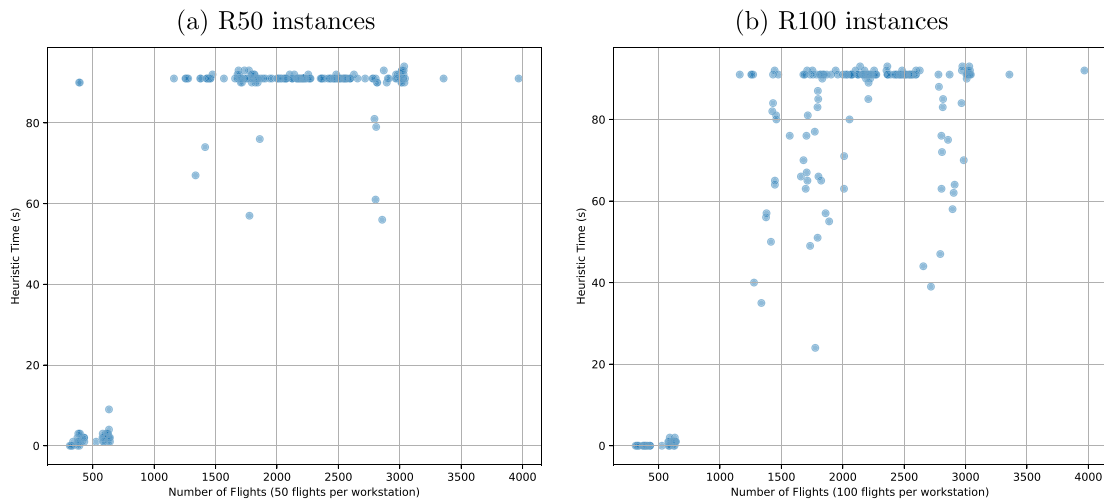


Fig. 11. Solution time versus instance size.

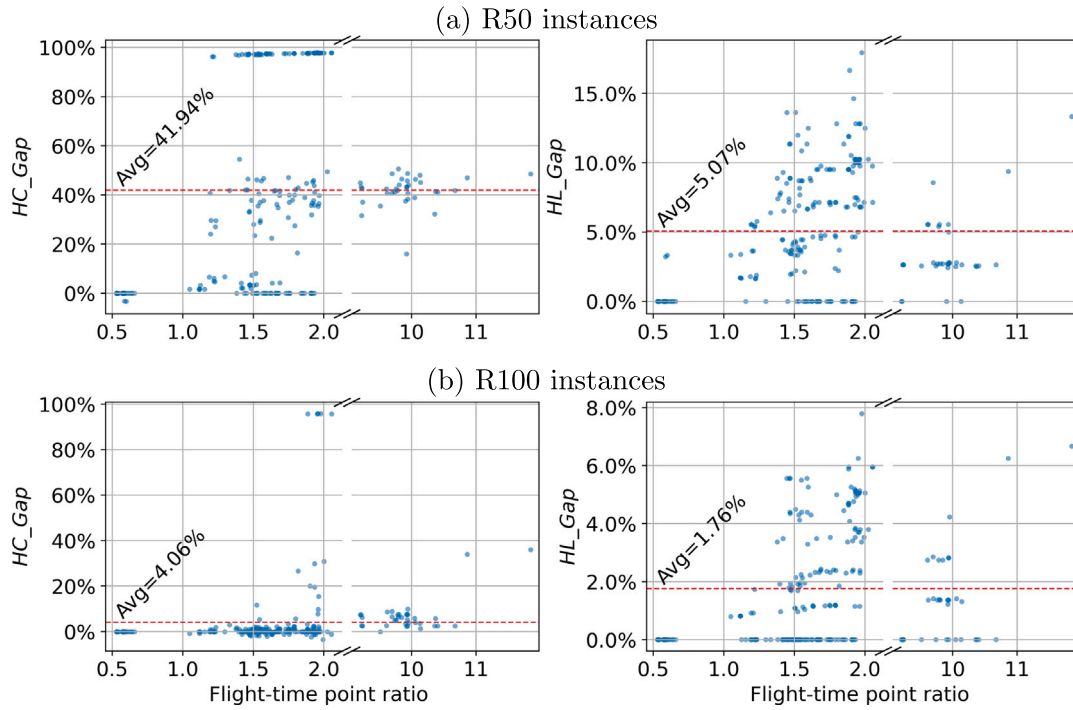


Fig. 12. Summary of the FSI-GA performance over varying flight to time-point ratio $\frac{|I|}{|T|}$.

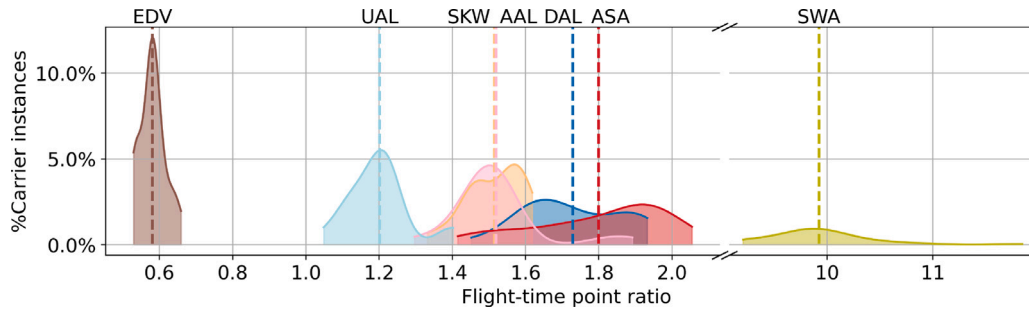


Fig. 13. Distribution of flight to time-point ratio among instances of each carrier.

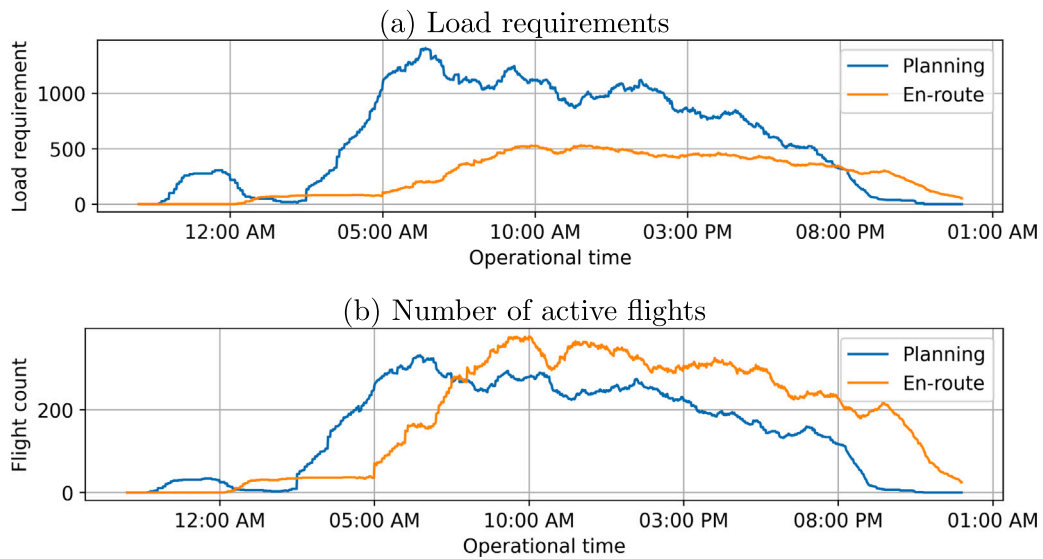


Fig. 14. Planning and en-route phases over the operational hours in the problem instance of Delta Air Lines 7 Sep 2019.

Table 2

Computational performance of the FSI-GA solution for (PWP) compared to the Cplex incumbent solution and Lagrangian lower bound for R50 instances.

Instance	Count	FSI-GA vs. Lag		FSI-GA vs. Cplex			Cplex		Time (s)	
		<i>HL_Gap</i>	Solved	<i>HC_Gap</i>	C+	H+	<i>C_Gap</i>	Solved	CTime ^a	HTime ^b
6	6	0.0	100	0.0	–	–	0.0	100	1.3	0.2
7	13	0.5	84.6	0.5	15.4	–	0.0	100	10.8	15.1
8	5	0.0	100	0.0	–	–	0.0	100	4.0	1.6
10	1	0.0	100	0.0	–	–	0.0	100	5.0	1.0
11	6	0.0	100	0.0	–	–	0.0	100	9.0	1.8
12	17	0.0	100	0.0	–	–	0.0	100	15.5	2.8
23	1	3.3	–	–1.6	–	100	4.8	–	3601.0	91.0
25	4	2.1	–	–2.0	–	100	4.0	–	3600.8	91.0
26	1	0.0	100	–1.6	–	100	1.6	–	3601.0	67.0
27	2	2.7	–	–18.1	–	100	20.1	–	3601.5	91.0
28	7	3.4	14.3	–26.2	–	100	28.3	–	3601.3	88.6
29	3	4.3	–	–50.9	–	100	52.7	–	3602.0	91.3
31	1	7.4	–	0.0	–	–	6.9	–	3601.0	91.0
33	6	4.2	–	–15.7	–	83.3	19.2	–	3605.2	91.7
34	7	4.6	–	–35.1	–	100	38.0	–	3610.4	91.0
35	8	5.0	12.5	–27.4	–	62.5	30.7	–	3608.4	87.1
36	11	4.6	–	–47.2	–	90.9	49.7	–	3607.1	91.1
37	3	4.7	33.3	–34.8	–	100	37.1	–	3604.3	86.0
38	2	7.9	–	–50.6	–	100	53.8	–	3606.5	91.0
39	1	2.0	–	–33.3	–	100	35.7	–	3607.0	91.0
40	6	6.5	–	–50.5	–	100	53.1	–	3607.5	91.0
41	9	9.7	–	–64.0	–	100	66.9	–	3609.0	91.0
42	5	10.0	–	–72.6	–	100	75.2	–	3605.8	91.4
43	11	7.0	–	–75.3	–	100	76.4	–	3610.4	91.0
44	10	6.8	–	–67.1	–	100	69.5	–	3610.3	91.0
45	5	8.5	–	–51.9	–	100	56.1	–	3612.4	91.2
47	7	11.1	–	–97.6	–	100	97.9	–	3619.1	91.0
48	4	8.0	–	–37.0	–	100	42.4	–	3616.8	91.0
49	9	9.7	–	–59.7	–	100	63.7	–	3616.0	91.1
50	13	9.1	–	–84.5	–	100	85.9	–	3617.1	91.0
51	11	9.4	–	–87.7	–	100	89.0	–	3614.8	91.0
52	1	17.9	–	–97.7	–	100	98.1	–	3604.0	92.0
53	1	2.6	–	–35.5	–	100	37.1	–	3602.0	91.0
54	1	2.6	–	–41.2	–	100	43.2	–	3603.0	91.0
55	3	2.5	33.3	–34.1	–	100	35.6	–	3601.3	87.7
56	5	1.6	40.0	–40.2	–	100	45.1	–	3603.2	82.2
57	3	1.7	33.3	–44.9	–	100	48.3	–	3604.3	79.7
58	2	2.6	–	–36.7	–	100	40.8	–	3603.5	91.0
59	4	4.1	–	–40.5	–	100	49.1	–	3605.0	91.5
60	14	3.7	–	–44.0	–	100	49.5	–	3605.3	91.6
67	1	9.4	–	–47.0	–	100	52.8	–	3602.0	91.0
79	1	13.3	–	–48.5	–	100	55.6	–	3604.0	91.0
Overall	231	5.1	23.4	–41.9	0.9	76.6	44.3	20.8	2861.1	72.6

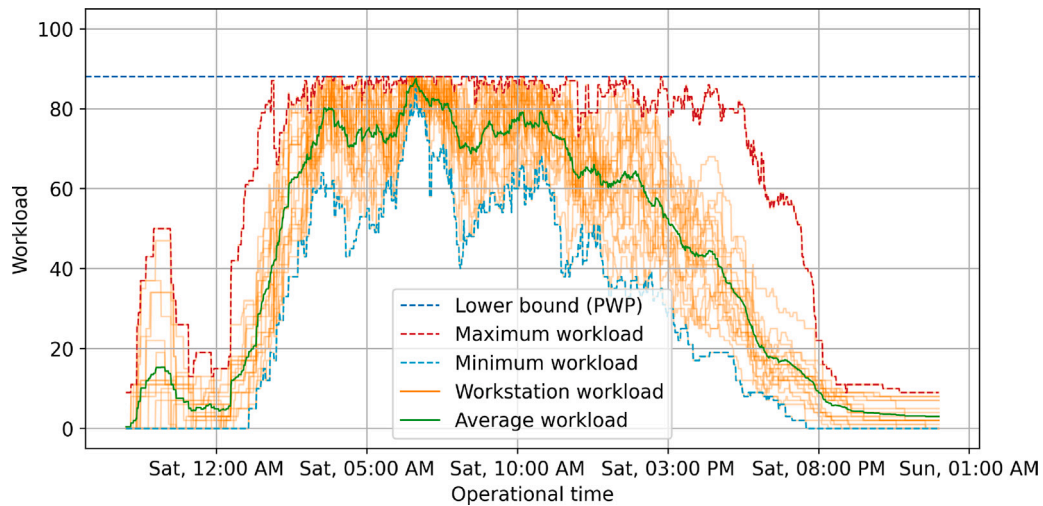
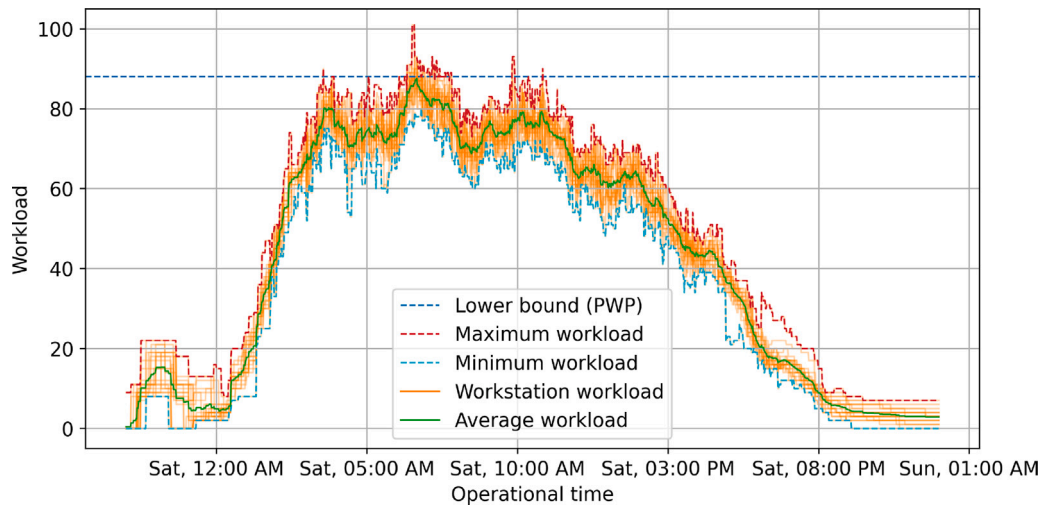
^a Heuristic runtime in seconds.^b Solver runtime in seconds (clock time).**Fig. 15.** Distribution of workload in the optimal solution for (PWP) of R100 Delta Air Lines 7 Sep 2019 instance.

Table 3

Computational performance of the FSI-GA solution for (PWP) compared to the Cplex incumbent solution and Lagrangian lower bound for R100 instances.

Instance		FSI-GA vs. Lag		FSI-GA vs. Cplex			Cplex		Time (s)	
K	Count	HL_Gap	Solved	HC_Gap	C+	H+	C_Gap	Solved	CTime ^a	HTime ^b
3	19	0.0	100	0.0	–	–	0.0	100	0.4	0.0
4	5	0.0	100	0.0	–	–	0.0	100	1.0	0.0
5	7	0.0	100	0.0	–	–	0.0	100	1.0	0.6
6	17	0.0	100	0.0	–	–	0.0	100	1.2	0.9
11	1	0.8	–	0.8	100	–	0.0	100	1261.0	91.0
12	4	0.6	25.0	0.2	25.0	–	0.4	50.0	2347.0	78.2
13	3	0.0	100	0.0	–	–	0.0	100	376.0	49.3
14	10	0.4	70.0	–0.4	–	30.0	0.7	60.0	2005.4	78.0
15	1	0.0	100	–1.8	–	100	1.8	–	3601.0	76.0
16	6	0.9	50.0	0.6	33.3	–	0.3	83.3	1894.7	78.7
17	15	0.6	66.7	–0.9	6.7	20.0	1.5	60.0	2112.3	74.5
18	14	0.9	42.9	–1.0	7.1	64.3	1.9	21.4	3143.4	81.9
19	3	2.1	–	–0.3	33.3	66.7	2.4	33.3	2575.7	91.3
20	15	2.8	20.0	–0.0	33.3	26.7	2.7	20.0	3108.3	87.1
21	16	3.4	6.2	–0.1	25.0	37.5	3.4	12.5	3272.6	91.2
22	15	1.9	13.3	–3.2	6.7	86.7	5.0	–	3608.5	90.5
23	7	4.7	–	–0.9	–	57.1	5.4	–	3604.1	91.4
24	13	3.4	–	–0.3	15.4	38.5	3.6	7.7	3386.5	91.1
25	24	4.6	–	–21.4	16.7	54.2	24.6	4.2	3528.0	91.0
26	2	3.9	50.0	–47.9	–	50.0	48.1	50.0	2170.0	68.0
27	4	0.3	75.0	–3.3	–	100	3.6	–	3600.8	66.2
28	8	0.2	87.5	–5.0	–	100	5.2	–	3602.8	75.4
29	6	0.7	66.7	–3.5	–	100	4.2	–	3601.2	77.5
30	14	2.1	–	–6.7	–	100	8.6	–	3601.7	91.3
33	1	6.2	–	–34.0	–	100	37.9	–	3602.0	91.0
39	1	6.7	–	–36.0	–	100	40.7	–	3602.0	92.0
Overall	231	1.8	43.3	–4.1	10.0	42.4	5.6	37.2	2442.0	67.1

^a Heuristic runtime in seconds.^b Solver runtime in seconds (clock time).**Fig. 16.** Distribution of workload in the heuristic solution for (ADP) of Delta Air Lines 7 Sep 2019 instance.

6.1. Flight data and workload determination

We generate test instances from U.S. historical domestic flight data of the Bureau of Transportation Statistics (Bureau of Transportation Statistics, 2023b) for the year 2019. Since the workload balancing problem is a daily problem and is airline specific, we form instances by considering flights departing on the same day and operated by the same carrier on a single day, excluding cancelled flights. A flight is defined by the departing and arrival airports and times. As flights may span multiple time zones, we standardize local flight times to Central Standard Time (CST, UTC -6). This ensures consistency when accounting for workload as a dispatcher center may oversee flights from

several time zones. For example, a flight departing from Atlanta at 10 am and another departing from Denver at 8 am, in local times, would need their planning tasks done in parallel. We generate 231 instances that span multiple airlines and varying sizes in terms of number of flights. Table 1 and Fig. 8 provides summary statistics of the instances.

There are relatively more instances from Delta Air Lines and Southwest Airlines as they are both among the top three carriers in the U.S. domestic passenger market in terms of market share (T4, 2022; Bureau of Transportation Statistics, 2023a), and have the largest problem instances, with the largest instance having 3968 flights. The time-points refer to the set T defined in Section 3. The number of time-points per instance is on average about half the number of flights, which implies

Table 4
Comparison of (PWP) and (ADP) solutions for R50.

Instance		Quality indicator ratio of (PWP)/(ADP)				
K	Count	Max. range	Avg. range	St.dev. range	Peak workload	Abs. workload dev.
6	6	2.47	2.68	2.68	0.93	2.76
7	13	2.43	2.37	2.65	0.92	2.57
8	5	2.26	2.35	2.77	0.90	2.55
10	1	1.72	2.28	2.61	0.81	2.50
11	6	1.87	1.97	2.10	0.86	2.12
12	17	1.66	1.96	1.98	0.83	2.10
23	1	1.86	1.62	2.02	0.89	1.91
25	4	1.90	1.66	2.27	0.86	1.97
26	1	1.88	1.74	2.42	0.82	2.04
27	2	1.91	1.52	2.23	0.85	1.85
28	7	1.69	1.56	2.19	0.88	1.90
29	3	1.68	1.45	2.11	0.85	1.84
31	1	1.87	1.80	2.34	0.85	2.09
33	6	1.71	1.76	2.13	0.87	2.13
34	7	1.70	1.73	2.18	0.85	2.09
35	8	1.64	1.70	2.04	0.84	2.04
36	11	1.65	1.60	2.08	0.82	1.92
37	3	2.14	2.33	2.59	0.81	4.95
38	2	1.64	1.60	2.01	0.80	1.89
39	1	1.72	1.63	2.69	0.81	1.93
40	6	1.45	1.50	1.79	0.84	1.87
41	9	1.47	1.51	1.89	0.84	1.86
42	5	1.52	1.56	2.02	0.86	1.89
43	11	1.55	1.55	1.92	0.83	1.88
44	10	1.53	1.46	1.87	0.83	1.86
45	5	1.35	1.45	1.63	0.83	1.84
47	7	1.37	1.39	1.82	0.82	1.70
48	4	1.32	1.34	1.54	0.81	1.74
49	9	1.45	1.40	1.68	0.83	1.78
50	13	1.23	1.28	1.43	0.82	1.64
51	11	1.36	1.27	1.54	0.81	1.61
52	1	1.50	1.31	1.80	0.92	1.58
53	1	1.54	1.35	1.58	0.82	1.63
54	1	1.39	1.38	1.58	0.83	1.67
55	3	1.39	1.44	1.57	0.77	1.69
56	5	1.43	1.39	1.54	0.80	1.62
57	3	1.34	1.29	1.67	0.63	2.17
58	2	1.52	1.38	1.65	0.82	1.63
59	4	1.44	1.30	1.40	0.77	1.54
60	14	1.34	1.27	1.43	0.77	1.51
67	1	1.17	1.16	1.25	0.76	1.47
79	1	1.17	1.09	1.12	0.77	1.37
Overall	231	1.62	1.60	1.95	0.83	1.97

that usually multiple flights have a load change at some time-point. Both the number of flights and number of time-points, along with the number of workstations, determine the size of the optimization models solved in terms of the number of decision variables and constraints. For an instance with $|I| = 1826$ flights, $|T| = 943$ time-points, and $|K| = 100$ workstations, problem (PWP) has 182,600 binary variables, and 96,126 constraints.

Determining flight load requirements is critical to the load balancing problem. As discussed earlier, it is commonly agreed that the load of a flight has two phases: planning and en-route. The planning phase includes tasks to get the flight plans ready, such as fuel planning and preparing the dispatch release. It starts 150 min before departure and lasts 120 min. There is no load in the 30 min interval before departure as the flight prepares for takeoff. The en-route phase starts at takeoff and ends at landing and includes monitoring the flight and the weather and dealing with any alerts. To put numbers to the load associated with the tasks in each phase, we base the load on flight distance. The idea is that longer flights require more preparation before takeoff and has larger meteorological regions to check while in the air, which serves as an indication of the load required from the dispatcher. Moreover, weather information at the arrival airport is not as reliable on longer flights due to the larger time span, which presents a higher risk of contingencies that may require revising the flight plan. In rare

case that may entail diversion to an alternate airport which would cause substantial workload on the dispatcher (Bartels, 2019; Collier and Bartels, 2021, 2023). We acknowledge that an accurate numerical representation requires data from actual flight dispatcher operations. However, no such study has been done in the literature, and the subjective definition of what constitutes workload presents another research problem by itself. We also note that unforeseen events such as abrupt changes in weather conditions are not captured explicitly in the workload. This is due to the difficulty in determining the timing of such random events as we account for workload at time-points when workload changes. It is implicitly aggregated in the workload corresponding to en-route flight monitoring.

The planning load and the en-route load, respectively, are assigned $\left\lceil \frac{\text{Distance (mi)}}{250} \right\rceil$ and $\left\lceil \frac{\text{Distance (mi)}}{1250} \right\rceil$ units of load per minute, adding one planning and en-route load level for every 250 and 1250 miles flown. The reasoning for using these intervals is related to the average distance covered by flight between weather forecast updates, which roughly equates to 1250 miles considering landing and take-off. Fig. 9 presents the distribution of load requirements over the 24-h operational cycle. Fig. 10 illustrates the distribution of flight planning and en-route loads per carrier over all testing instances.

The number of workstations in the operations centers of carriers is typically kept confidential and not shared with the public. According

Table 5
Comparison of (PWP) and (ADP) solutions for R100.

Instance		Quality indicator ratio of (PWP)/(ADP)				
K	Count	Max. range	Avg. range	St.dev. range	Peak workload	Abs. workload dev.
3	19	3.25	3.46	3.56	0.97	3.53
4	5	2.61	3.10	3.39	0.96	3.21
5	7	2.32	2.80	2.73	0.94	2.97
6	17	2.19	2.58	2.57	0.93	2.73
11	1	3.61	2.72	3.90	0.88	2.90
12	4	3.01	2.58	3.47	0.93	2.68
13	3	2.87	2.40	3.38	0.90	2.59
14	10	2.99	2.41	3.72	1.05	2.51
15	1	2.65	2.72	3.71	0.90	2.67
16	6	2.82	2.64	3.47	0.91	2.84
17	15	2.92	2.62	3.82	0.97	2.76
18	14	2.55	2.51	3.53	0.90	2.79
19	3	2.38	2.32	3.47	0.92	2.62
20	15	2.46	2.22	3.35	0.90	2.48
21	16	2.40	2.24	3.38	0.90	2.50
22	15	2.37	2.11	3.13	0.90	2.47
23	7	2.50	2.02	3.57	0.90	2.26
24	13	2.17	1.96	2.91	0.90	2.32
25	24	2.22	1.83	2.74	0.91	2.14
26	2	2.08	1.95	2.63	0.91	2.11
27	4	2.21	1.90	2.39	0.84	2.09
28	8	2.26	1.89	2.43	0.86	2.08
29	6	2.18	1.82	2.38	0.85	1.98
30	14	2.06	1.68	2.23	0.85	1.84
33	1	1.73	1.55	2.08	0.82	1.71
39	1	1.97	1.47	1.98	0.89	1.61
Overall	231	2.49	2.29	3.07	0.91	2.48

to statistics reported by Hope (2017), Air Canada utilizes 22 flight dispatchers working in 8-h shifts, managing an average of 750 flights daily. With all workstations operating 24 h and three dispatcher shifts distributed throughout the day, each workstation handles an average of 102 flights. For the sake of simplicity, we round this number to 100 flights per workstation and use it to determine the number of workstations as $\lceil \frac{|I|}{100} \rceil$. Additionally, we test a rate of 50 flights per workstation, and each instance is tested at ratios resulting in two classes of instances: R100 and R50 instances refer to an average of 100 and 50 flights per workstation, respectively. A rate of 50 flights leads to about double the number of workstations for the same number of flights, which increases the solution space and is expected to lead to more difficult instances than R100. Note that we do not enforce that 100 (or 50) flights are assigned to each workstation, but rather we use it to set the number of workstations only.

6.2. Quality of solutions

In this section, we assess the quality of the solutions generated by FSI-GA relative to the Lagrangian lower bound and to the performance of the commercial solver Cplex. The genetic algorithm is applied with a population size of 128, elite count of 16, mutation rate of $\frac{1.5}{|I|}$, crossover rate of $\frac{3.5}{|I|}$, initial time-point set size $\tau = 20 + \max \left\{ 4, \left\lfloor \frac{|I|}{10} \right\rfloor \right\}$, and a time limit of 90 s for the full heuristic. Cplex is applied with a time limit of 1 h clock time (TiLim = 3600).

Tables 2 and 3 report on the gaps between Cplex and FSI-GA and Lagrangian bound, the percentage of instances where either method outperforms the other, and on computational times for R100 and R50 instances. We report the results by averaging instances with the same number of workstations K . The Cplex MIP Gap is calculated as $C_Gap = 100 \times \frac{\text{Cplex Incumbent} - \text{Cplex lower bound}}{\text{Cplex Incumbent}}$. The gap between Cplex and FSI-GA is calculated as $HC_Gap = 100 \times \frac{\text{FSI-GA solution} - \text{Cplex Incumbent}}{\text{Cplex Incumbent}}$. Similarly, the gap between FSI-GA and the Lagrangian bound is calculated as

$HL_Gap = 100 \times \frac{\text{FSI-GA solution} - z_{MP}}{z_{MP}}$. Columns C+ and H+, respectively, contain the proportion of instances where Cplex incumbent outperforms the FSI-GA solution and vice versa. Column HL_Gap contains the difference between FSI-GA objective and the Lagrangian bound, relative to the Lagrangian bound. Finally, column Solved under FSI-GA vs. Lag contains the proportion of instances where FSI-GA has reached the Lagrangian bound.

The Lagrangian bound is very tight as it is confirmed to be equal to the optimal objective value in all problem instances with known optimal solutions, which corresponds to 24.2% and 48.5% of the R50 and R100 instances. In the remaining instances the gaps are 6.7% and 3.1% for R50 and R100 respectively. Cplex fails to solve 78.9% and 60.6% of the R50 and R100 instances, respectively. It is only able to solve smaller instances with fewer flights and number of workstations. The average gaps are 44.3% and 5.6% and the average CPU times are 2861 s and 2442 s for R50 and R100. Extending the time limit to 3 h was not sufficient for solver to obtain an optimal solution, it failed to improve the MIP gap for both R50 and R100 in two instances tested. When compared to the Lagrangian gap and Cplex, FSI-GA demonstrates robust performance across different settings. It outperforms Cplex in solution quality and finds solutions that are very close to the Lagrangian bound in less than 100 s. It surpasses the Cplex solver in 76.6% of R50 instances and 42.4% of R100 and obtains a better workload balance, outperforming the solver by an average of 41.9% and 4.1% in R50 and R100 instances, respectively.

For (ADP), the Cplex solver and FSI-GA are applied under the same settings used in the testing of (PWP). The Cplex solver failed to obtain a solution of comparable quality within the time limit and finished with a MIP gap of over 90% for the 10 smallest instances in the number of flights. Since Cplex failed to solve any of these instances, we did not solve the rest of the instances using Cplex for (ADP). On average, the heuristic objective outperforms the Cplex incumbent by 96% in R100 and 94% in R50 in these 10 instances.

Fig. 11 illustrates the relation between the problem size and the solution time. As expected, increasing the number of flights makes the

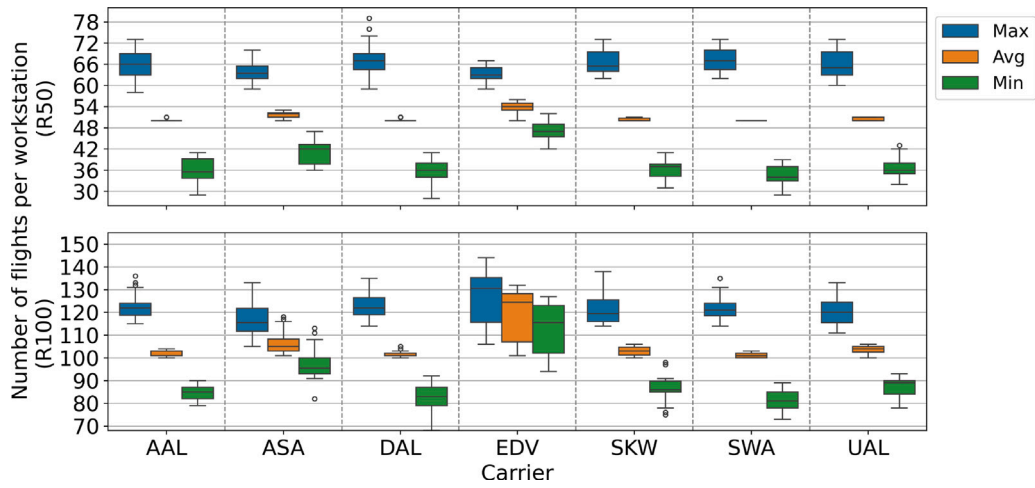


Fig. 17. Distribution of the number of flights per workstation for different carriers in the FSI-GA solutions for (PWP).

problem intractable more quickly in R50 instances than R100. This is evident as more instances terminating earlier than the set time limit. It is also seen that the heuristic still manages to obtain optimal solutions for instances with more than 1000 flights within a short time. These results confirm the effect of the number of desks on the difficulty of the instances. For the same instance, more desks leads to a larger solution space and a more difficult instance to solve.

Comparing R100 and R50, it is clear that Cplex performs much worse on R50 instances and on higher number of workstations and flights, the effects are not as pronounced on FSI-GA. time-points T is another factor that may impact the difficulty of the instances that is not covered in Tables 2 and 3. It should be noted that, in these tables, the $|K|$ column indicates the number of workstations, which is determined by the number of flights $|I|$ in the problem instance as $\lfloor \frac{|I|}{50} \rfloor$ for R50, and $\lfloor \frac{|I|}{100} \rfloor$ for R100. Therefore, as the number of flights in each instance differ, so does the number of instances for each $|K|$ value. In order to analyze the effects of time-points we further plot the gaps relative to the ratio of number of flights and time-points $\frac{|I|}{|T|}$ in Fig. 12.

There seems to be smaller variability in the heuristic solutions (when compared to the Lagrangian bound) for higher flight to time-point ratios. A similar trend is apparent for the Cplex gap. It is worth noting the jump in the ratio from 3 to 9. It turns out that SouthWest Airlines instances have a much higher flight to time-point ratio in the range of 9 to 11. This ratio ranges between 0.5 to 2.5 for the other carriers in the testing. This is probably due to the flight schedules operated by Southwest Airlines, where a high number of flights have the same departure/arrival times. In fact, the flight to time-point ratio seems to be a characteristic of the carrier, as illustrated in Fig. 13 which plots the distribution of flight to time-point ratios for each carrier. Endeavor airlines has lower ratios implying that few flights share the same time-points. A higher flight to time-point ratio may lead to fewer options for the improvement step in FSI-GA which is based on a two-swap and consequently fewer chances to move out of local optima.

6.3. Quality of the workload balance

The quality of workload balance achieved in the solution is influenced by the workload balance measure adopted. When minimizing the peak workload in (PWP), the objective value is set at time-points with high load requirements. This focus may lead to imbalance at other time-points, the absolute deviation measure in (ADP) addresses this imbalance as it minimizes the deviation from the average. For instance, Fig. 14 displays the distribution of total load requirements

in one instance corresponding to Delta Air Lines, September 7, 2019. Figs. 15 and 16 show, respectively, the distribution of workload among workstations in a (PWP) and an (ADP) solution with R100. The blue dashed line shows the Lagrangian lower bound, which is equal to the peak workload in this case. The red, cyan and green lines show the maximum, minimum and average workloads over all workstations throughout the day. These do not correspond to any single workstation but rather show the range over all workstations. The orange lines show individual workstation workloads. In a balanced workload, it is desirable to have all these lines in proximity.

As workload balance refers to the equal spread among all workstations so that no one is carrying excessive burden relative to the others, the range of workload variation among workstations should be the smallest possible throughout the work shift. As flights are assigned to one workstation, there will be workload variations at certain time points. Figs. 15 and 16 show that the absolute deviation measure can achieve this balance, while the peak workload measure succeeds in shaving off peak workload, but may lead to high variations in workstation load at non-peak times. This underscores the limitations of the peak workload balance measure and suggests that while it is effective in balancing workloads at peak times, it may not provide a uniformly balanced workload throughout the day. The absolute deviation measure can obtain a more balanced workload distribution, potentially with a higher overall peak workload. This is further supported by the results reported in Tables 4 and 5, which provide the comparative statistics of the peak workload and absolute deviation solutions over all instances. All statistics are calculated as the ratio of the value for the (PWP) solution, and that of the (ADP) solution. The (PWP) always results in smaller peak workload with a ratio of 0.83 on average. This comes at the expense of doubling the range on average compared to the (ADP) solution. Similarly, the (ADP) solutions lead to better workload balance range not only in the objective value but also in the maximum, average, and standard deviation.

We further analyze the balance in terms of number of flights and workload per workstation over all instances. Since these two statistics are variable over time, we plot the minimum, maximum and average distributions for each carrier using box plots in Figs. 17 and 18 for (PWP), and Figs. 19 and 20 for (ADP). Looking at the average flight box plots (Figs. 17 and 19 in orange), it is noticeable that there is little variation in the average number of flights among workstations. This is not surprising as the average is a characteristic of the instance and not the solution. These box plots do tell us, however, that each airline operates similar daily schedules. The maximum and minimum statistics are an outcome of the solution and their proximity to the average may

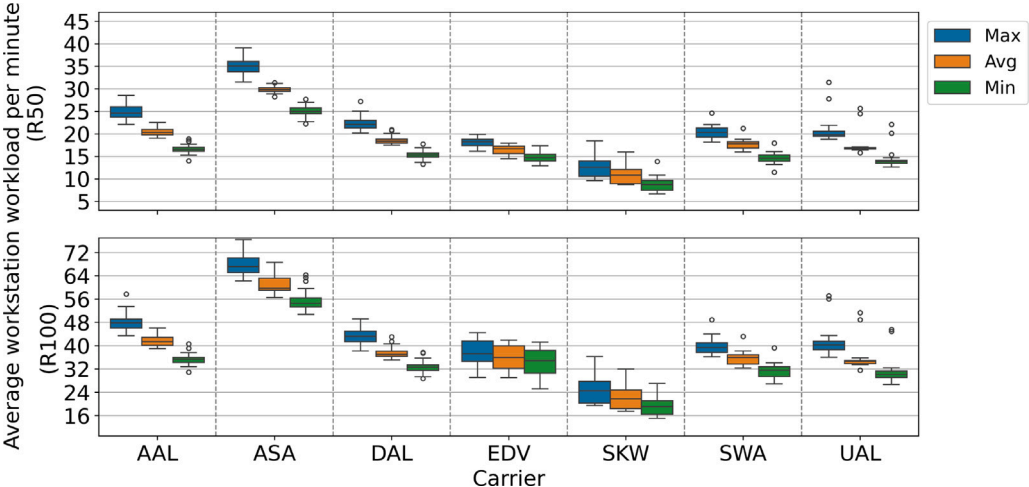


Fig. 18. Distribution of the workload for different carriers in the FSI-GA solutions for (PWP).

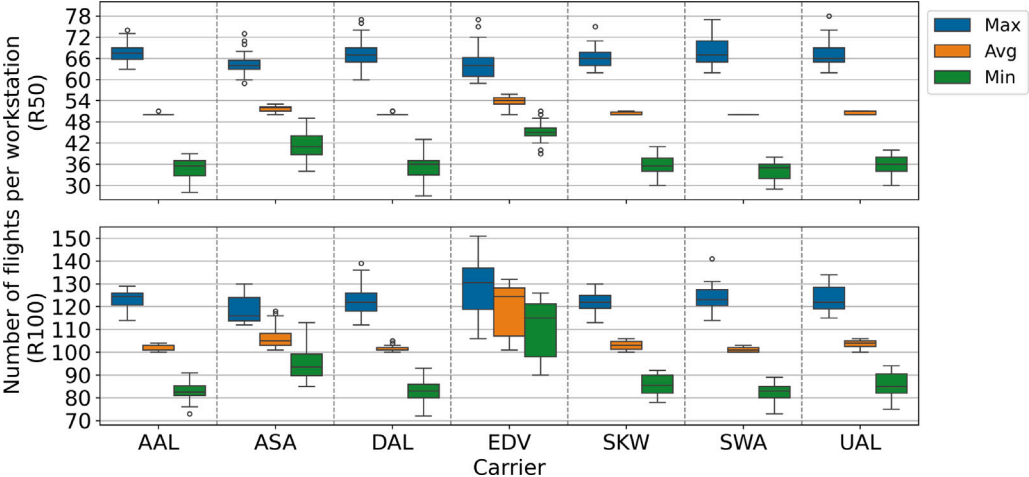


Fig. 19. Distribution of the number of flights per workstation for different carriers in the FSI-GA solutions for (ADP).

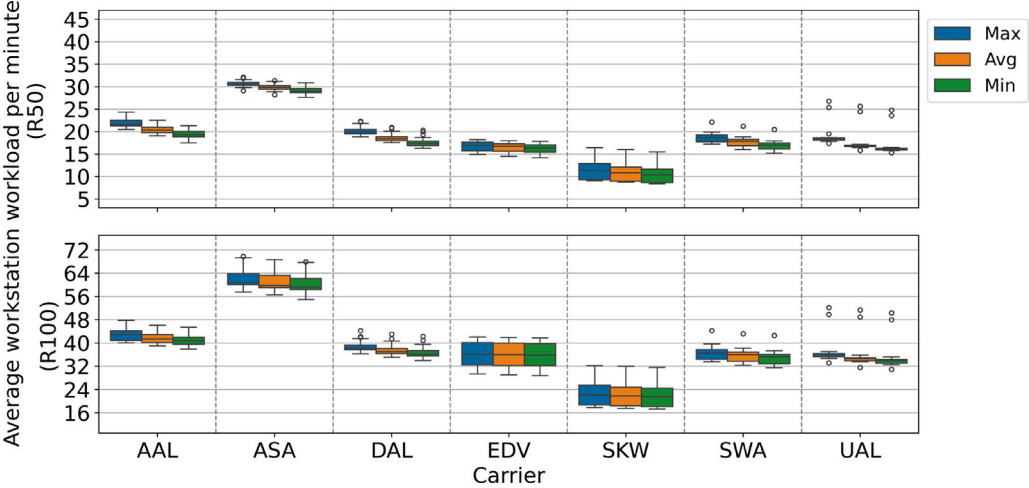


Fig. 20. Distribution of the workload for different carriers in the FSI-GA solutions for (ADP).

indicate whether the workstations in the same instance have similar number of flights over time. Except of Endeavor, all other solutions show some variability that we further analyze based on workload.

Similar to flights, the average workload is independent of the solution, as it is a function of the total load requirements of the flights and their schedule. On the other hand, high deviations of the minimum and maximum workloads from the average indicate that some workstations are being overloaded relative to others. R50 instances show more variability, which may be explained by the fact that there are about double the number of workstations compared to R100. The range in workload also depends on the magnitude of load. For example, on average, Alaska Airlines flights fly longer distances compared to others, which results in its workload values to be higher since the load is based on distance. Hence, higher workload from one airline to another is not an indicator of workload imbalance. It does highlight the limitation of how we set the number of workstations based on a preset ratio. While this may not be realistic for all operations, it does not limit the applicability of our models and methodology since the number of workstations is an input.

7. Conclusion

The objective in workload balancing for flight dispatchers is to assign flights to workstations in order to balance workload between them at any given time during the workday. The problem is unique due to multi-tasking, dynamic workload depending on flights status, multiple sources of information from different technologies, and interaction with various flight operations teams. To the best of our knowledge, this work is the first to study a realistic setting that models multi-tasking explicitly and introduces a load measure that is not measured in units of time. We developed mixed integer programming models with two nonlinear workload balance measures, derived a Lagrangian based lower bound, and built a fast focus-search-improve heuristic that finds high quality solutions for real world instances within seconds.

Numerical testing demonstrates the efficiency of the proposed approach in that the Lagrangian lower bound is very tight, and the heuristic finds proven optimal solutions in 33.4% of the instances and is on average 3.5% away from optimality. The direct solution using a commercial solver performed poorly on the peak workload objective and failed to solve even the smallest instances for the absolute deviation objective. Moreover, when assessing the quality of the peak workload and the absolute deviation objectives in terms of workload balance, we found that the latter achieves better workload balance throughout the day at the expense of slightly higher overall peak workload.

The current work opens the door to several new research extensions. Considering other workload balance measures, modeling additional practical features related to flight, dispatcher and workstation requirements, and embedding the proposed Lagrangian approach within an exact solution method that is capable of solving large sizes, are important directions. Moreover, the definition as well as the measure of workload deserves further study and would require experimental data through ethnography studies. As we focus on assigning flights to workstations in our work, there is opportunity to work on the follow up problem, which is to assignment of dispatchers to workstations. The latter is an important research question that needs to take into account dispatcher related aspects like union rules, and work preferences.

CRedit authorship contribution statement

Serkan Turhan: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Fatma Gzara:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Samir Elhedhli:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Appendix

Algorithm A.1: Focus-Search-Improve Genetic Algorithm

Input: Instance Data: I, K, T, h_{ii}

Time Limit

Extension Size for Partial Problem

Initial time-point set: T'

Output: Heuristic Solution

- 1 **Step 2:** Initialize best solution: all flights assigned to the first workstation.
- 2 **Step 3:** Initialize the genetic algorithm with T' .
- 3 **Step 4:** Iterate with the Focus-Search-Improve procedure until either the lower bound or the time limit is reached.
- 4 **while** Best objective on $T > \text{Lower Bound}$ **and** Elapsed Time $< \text{Time Limit}$ **do**
 - 5 **Step 4.1:** Iterate the genetic algorithm.
 - 6 **Step 4.2:** Update the best known solution with the genetic algorithm's last solution pool, evaluated on the full time-point set T .
 - 7 **Step 4.3:** If the genetic algorithm's best objective value on T' is less than the best known objective calculated on full set T , extend the time-point set T' according to Algorithm A.2 and extension size.
- 8 **Step 5:** Return the best known solution for the objective calculated on full set T .
- 9 **return** Best Solution on T .

Algorithm A.2: Focus Procedure of the Algorithm

Input: Instance Data: T, T'

Extension Size

Solutions from Algorithm A.3

Output: Updated set of times T'

- 1 **Step 1:** Select the solutions with best (minimum) objective values in the solutions from Algorithm A.3
- 2 **Step 2:** For each solution, calculate the workload balance error at each time-point not included in T' as the difference between the current objective value and the objective value when that time-point is included.
- 3 **Step 3:** Calculate the total workload balance error at each time-point as the sum of balance errors of individual solutions.
- 4 **Step 4:** If all time-points have zero workload balance errors, end the procedure and return the current T'
- 5 **Step 5:** Select a time-point with the highest total error, breaking ties arbitrarily.
- 6 **Step 6:** Add the selected time-point to the set T' .
- 7 **Step 7:** Reduce the Extension Size by 1. If becomes zero, return T' and end the procedure.
- 8 **Step 8:** Run the Search procedure in Algorithm A.3 for 2 iterations to tune the current population.
- 9 **Step 9:** Reapply the procedure from the beginning.

Data availability

Data will be made available on request.

Algorithm A.3: Search Procedure of the FSI-GA.

Input: Instance Data: I, K, T'
Algorithm Parameters: Iteration Limit, Population Size, C^{elite} , Best known objective on T
Output: Feasible Solutions improved for T'

- 1 **Step 1:** Initialize the population randomly.
- 2 **Step 2:** Iterate until the limit has been reached.
- 3 **for** $\text{Iteration} = 1$ **to** Iteration Limit **do**
- 4 **Step 2.1:** Calculate the population objective values on T' .
- 5 **Step 2.2:** Update the best solution found so far (on T'). If the best objective on T' is lower than best known objective on T , return the current population.
- 6 **Step 2.3:** Initialize the next population as empty.
- 7 **Step 2.4:** Select C^{elite} -many elites from the population and add them to the next population.
- 8 **Step 2.5:** Select parent pairs, then perform crossover and mutation on each pair to obtain the offspring.
- 9 **Step 2.6:** Apply the Improvement procedure in Algorithm A.4 to each offspring.
- 10 **Step 2.7:** Add the offspring to the next population. Replace the current population with the next population.
- 11 **Step 3:** Return the last population.
- 12 **return** Population.

Algorithm A.4: Improvement Procedure of the FSI-GA.

Input: Feasible Solution: Solution
Output: Improved Solution: Solution

- 1 **Step 1:** Iterate over all flight pairs.
- 2 **foreach** $\text{Flight } i \in \text{Solution}$ **do**
- 3 **foreach** $\text{Flight } j \in \text{Solution}$ **do**
- 4 **Step 2.1:** Evaluate the balance measure over time-points where either Flight i or j is active, store the balance value as *Current Imbalance*.
- 5 **Step 2.2:** Swap the workstation assignments of Flight i and j , evaluate the balance measure over the same set of time-points, store the balance value as *Swapped Imbalance*.
- 6 **Step 2.3:** If *Current Imbalance* is greater (worse) than *Swapped Imbalance*, accept the swap and apply the improvement procedure from the beginning with the new solution.
- 7 **Step 2.4:** If *Current Imbalance* is less than or equal to *Swapped Imbalance*, revert the swap and continue with the iterations.
- 8 **Step 3:** Return the solution only when no improvements are identified.
- 9 **return** Solution.

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